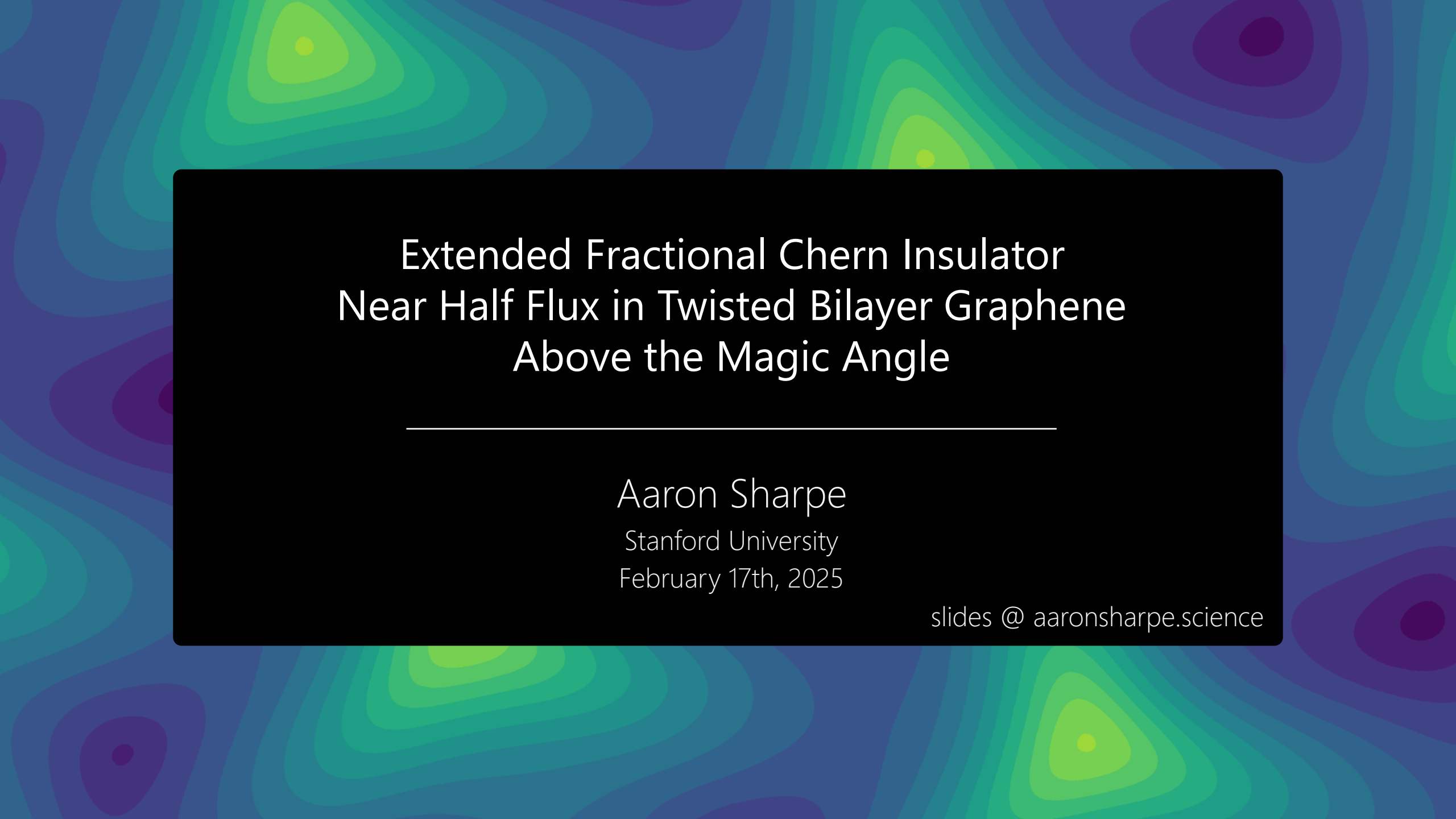




Extended Fractional Chern Insulator Near Half Flux in Twisted Bilayer Graphene Above the Magic Angle

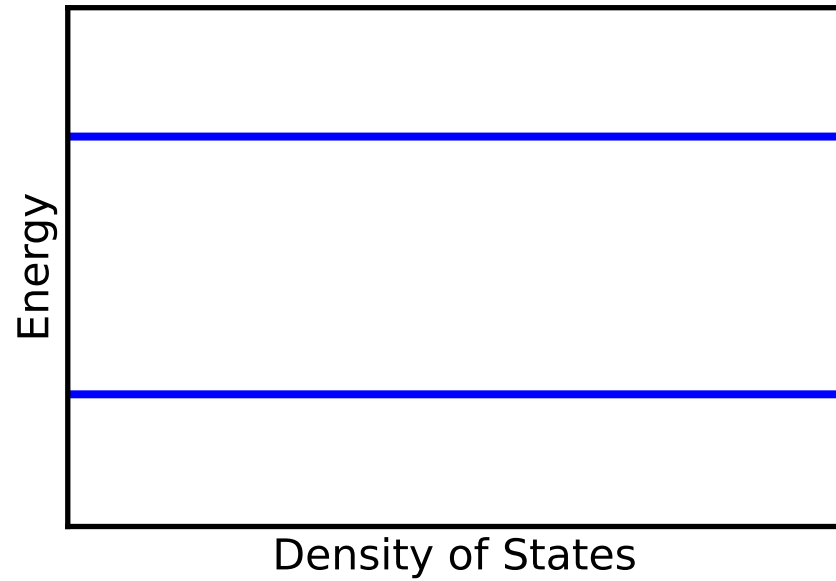
Aaron Sharpe

Stanford University

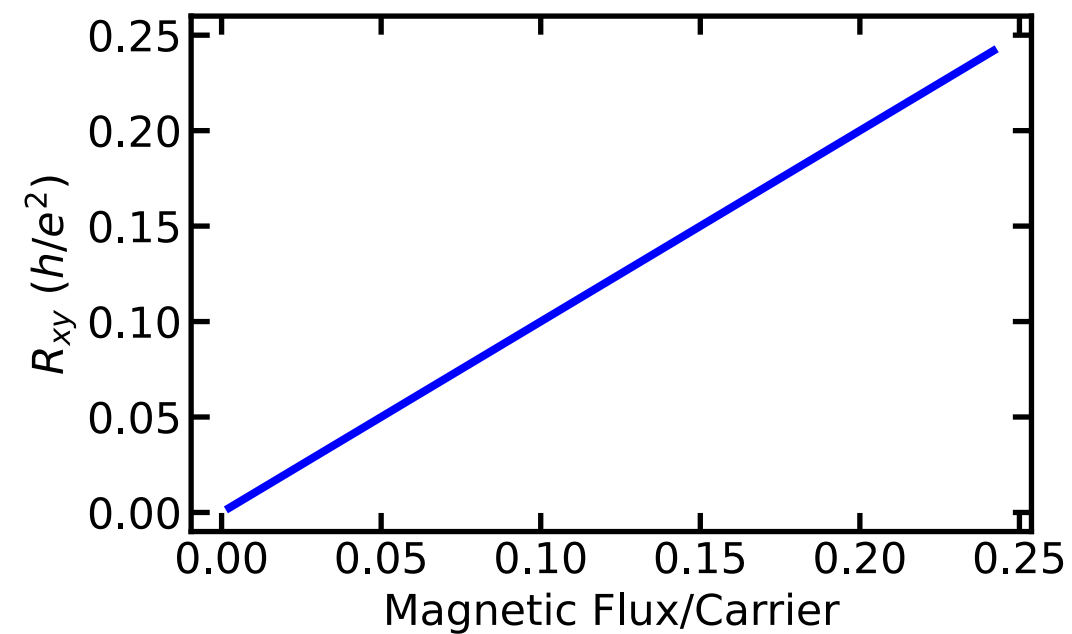
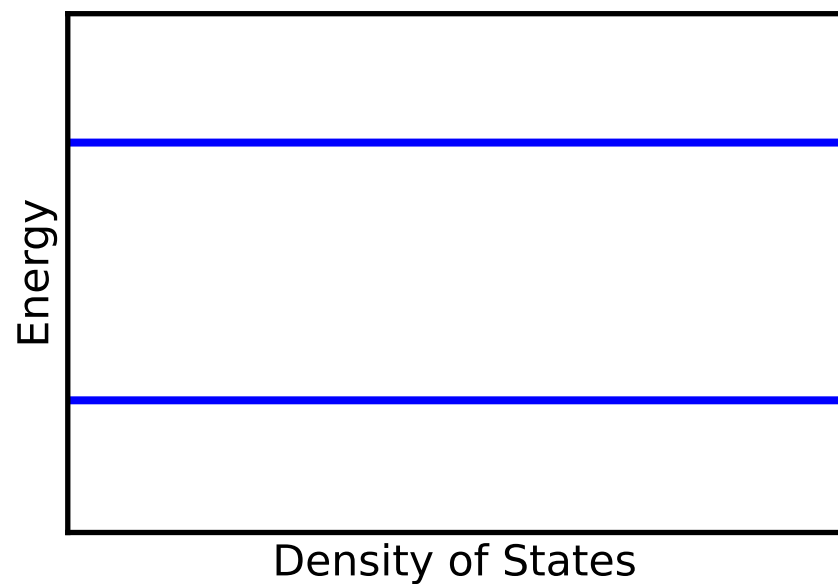
February 17th, 2025

slides @ aaronsharpe.science

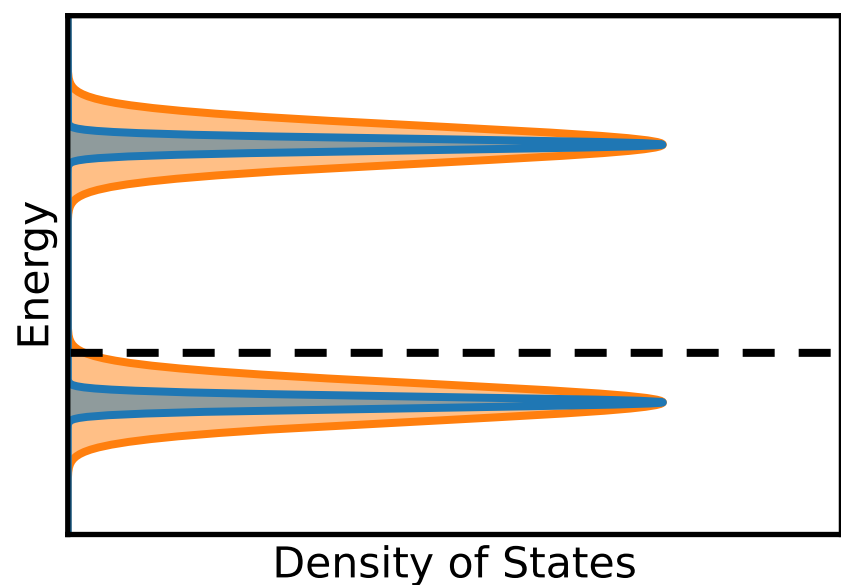
Landau Levels and Disorder



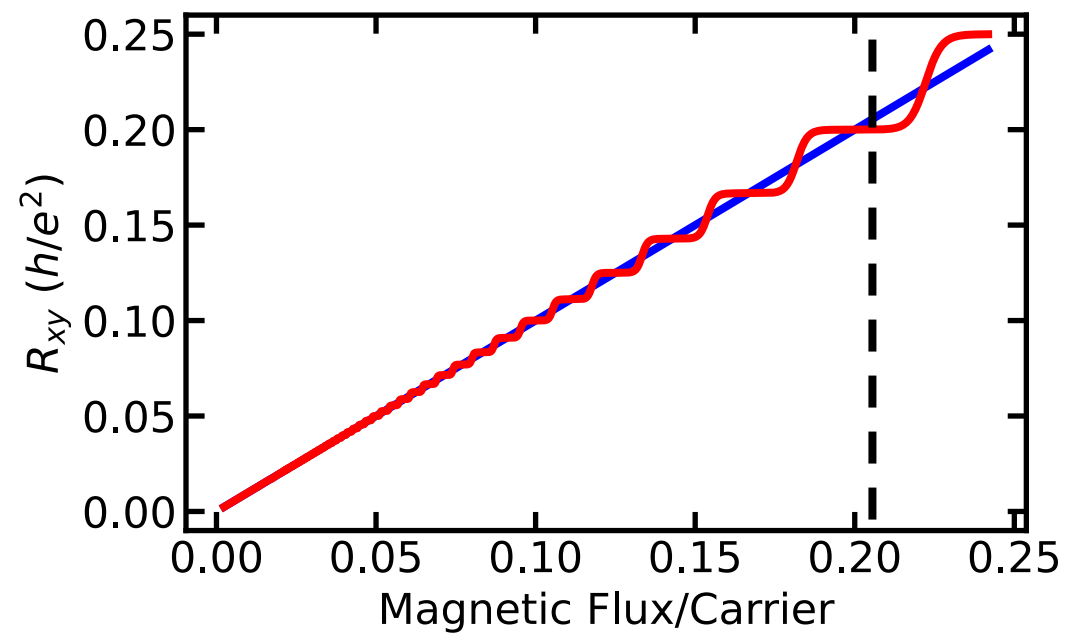
Landau Levels and Disorder



Landau Levels and Disorder

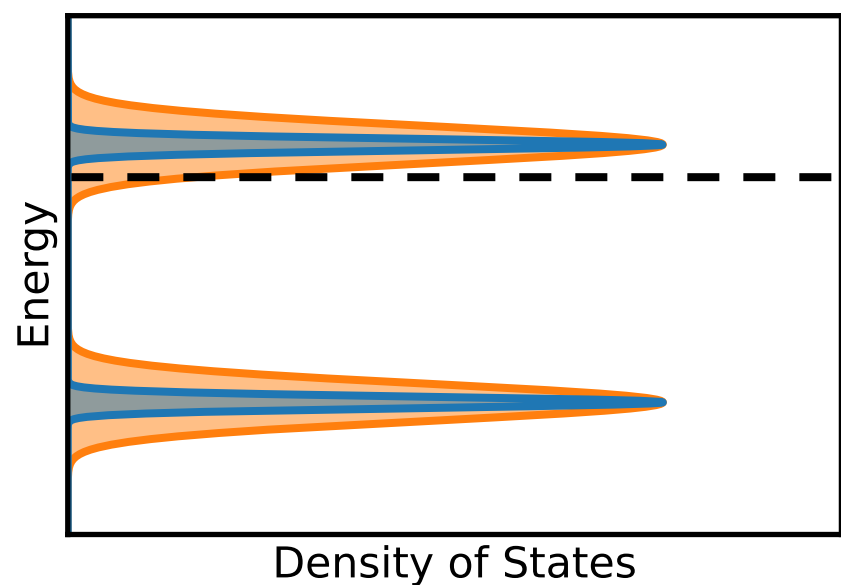


- Localized states
- Extended states

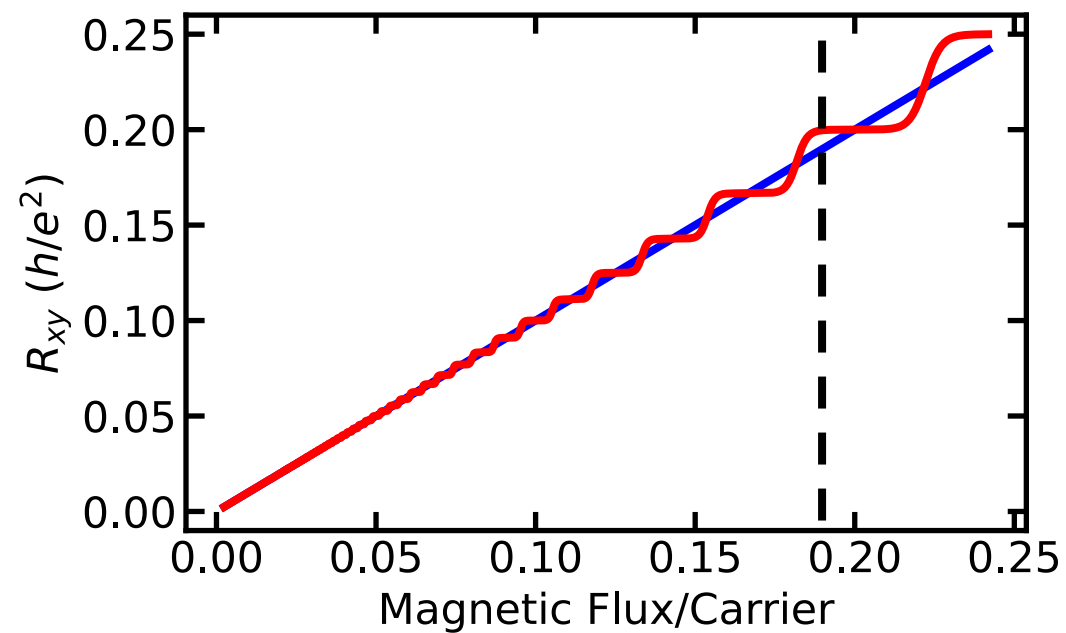


- Some disorder
- No disorder

Landau Levels and Disorder

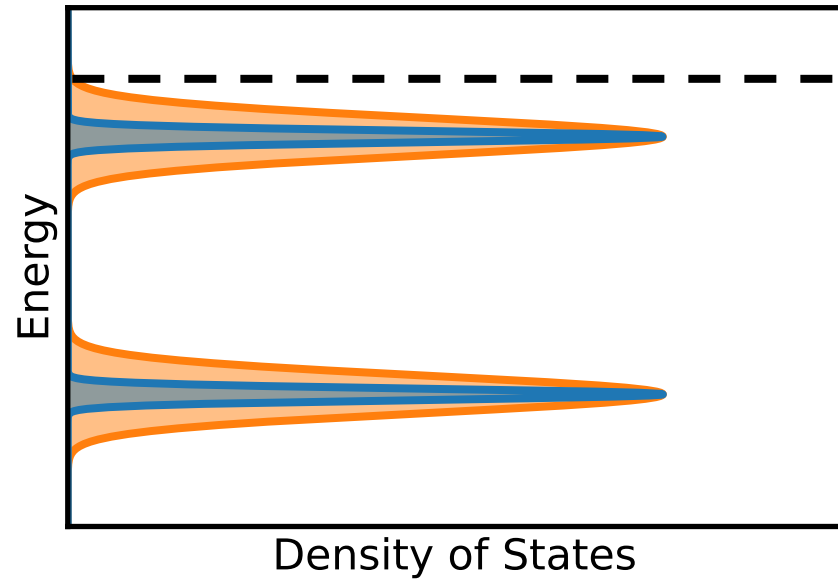


- Localized states
- Extended states

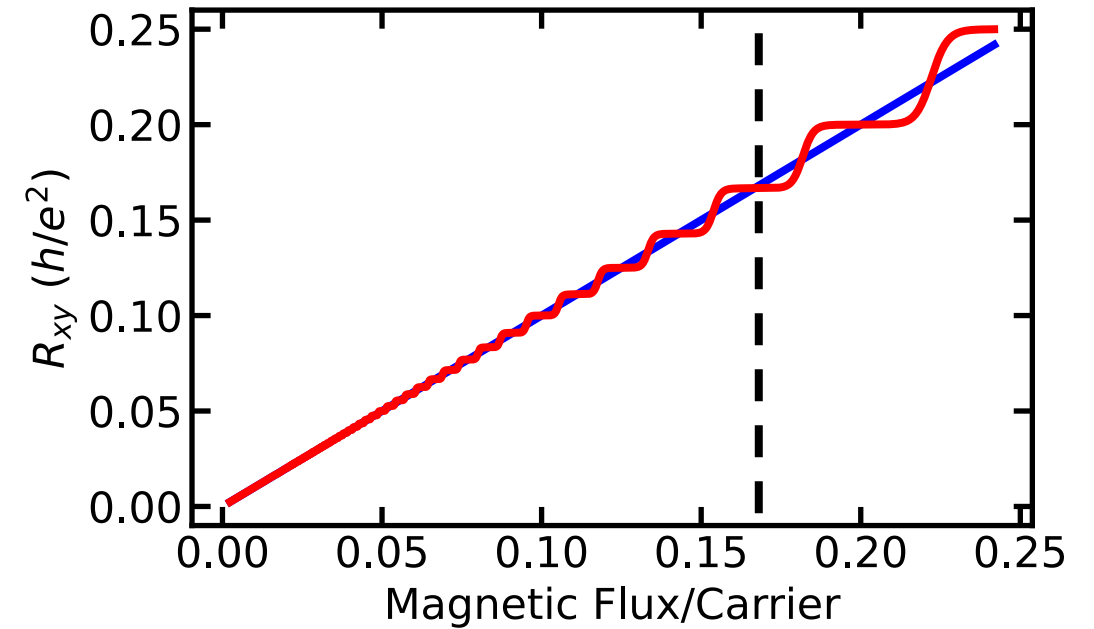


- Some disorder
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Landau Levels and Disorder



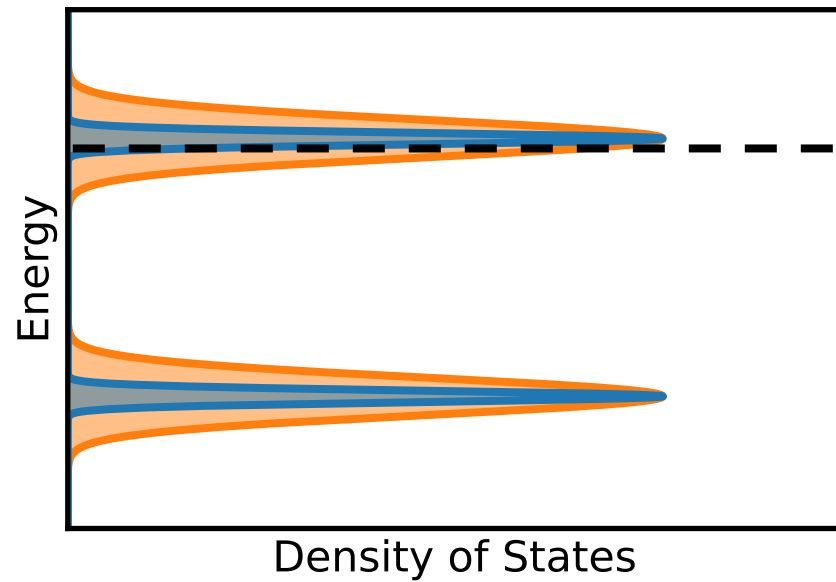
- Localized states
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Landau Levels and Disorder

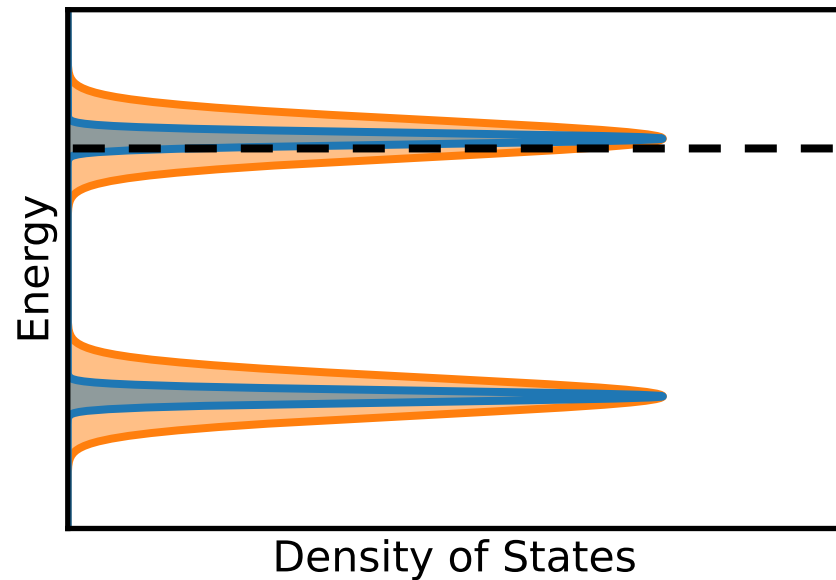
What can happen when doping into a LL?



- Localized states
- Extended states

Landau Levels and Disorder

What can happen when doping into a LL?



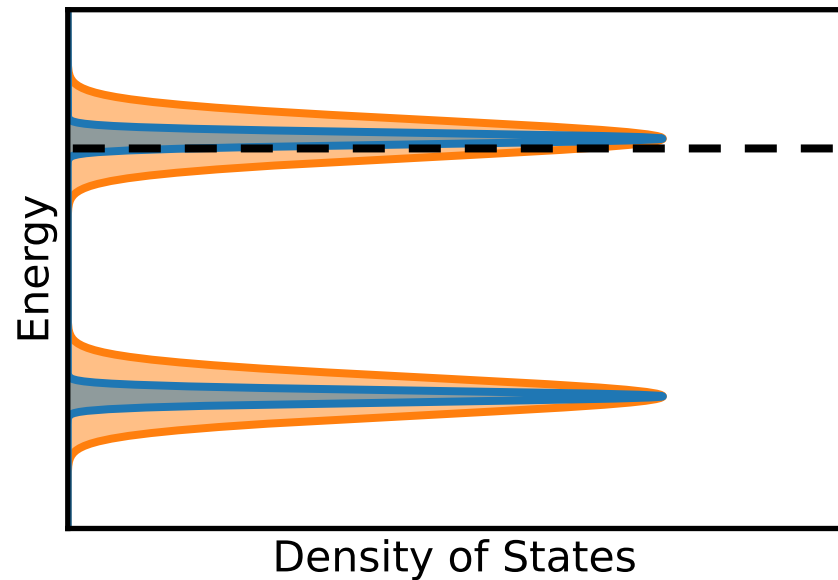
- Localized states
- Extended states

Metallic state:

- Finite R_{xx} and unquantized R_{xy}

Landau Levels and Disorder

What can happen when doping into a LL?



- Localized states
- Extended states

Metallic state:

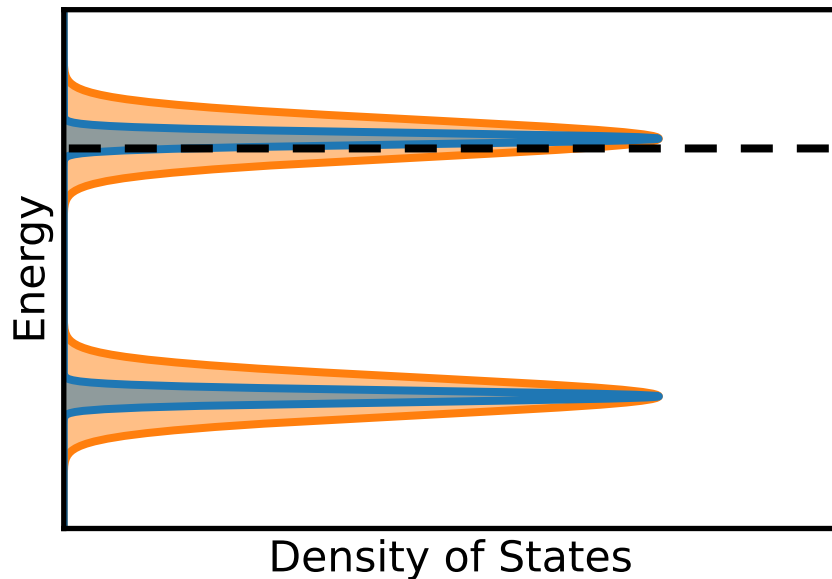
- Finite R_{xx} and unquantized R_{xy}

Incompressible FQH Liquid:

- At rational filling
- Zero R_{xx} and quantized R_{xy}
- Very sensitive to disorder

Landau Levels and Disorder

What can happen when doping into a LL?



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- Extended states

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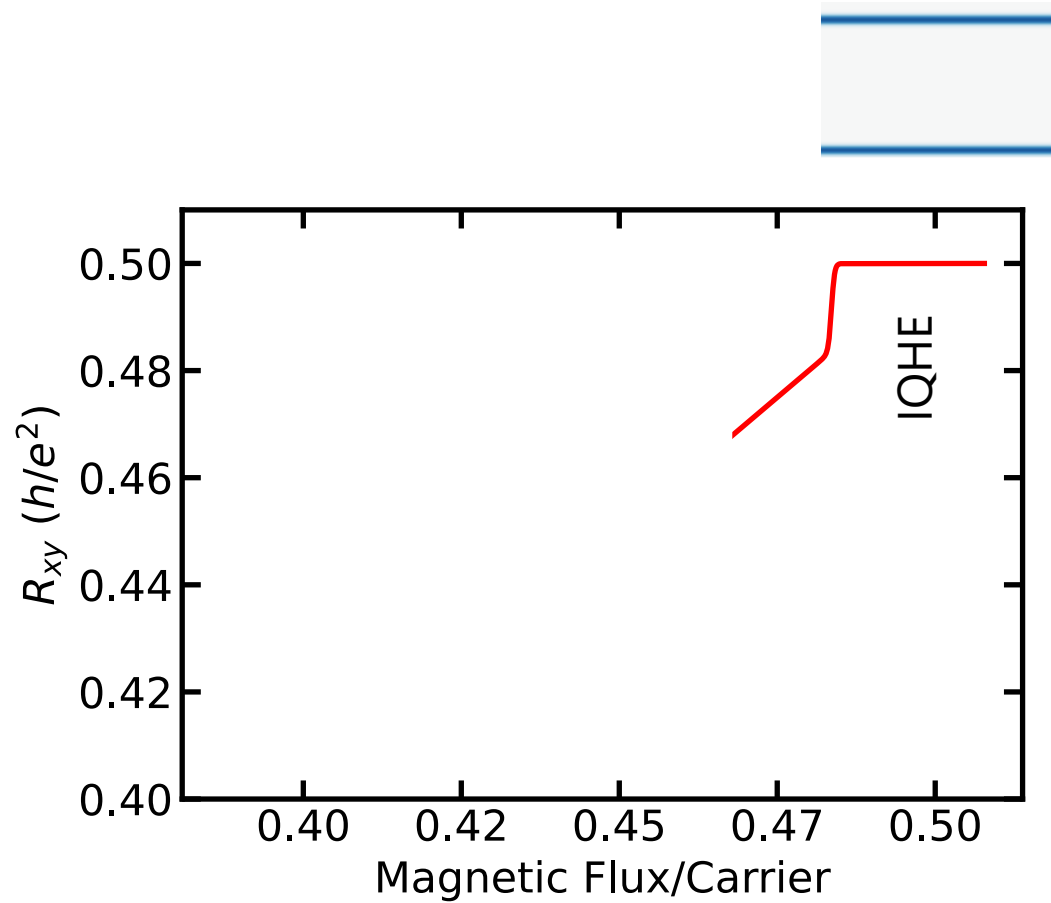
Incompressible FQH Liquid:

- At rational filling
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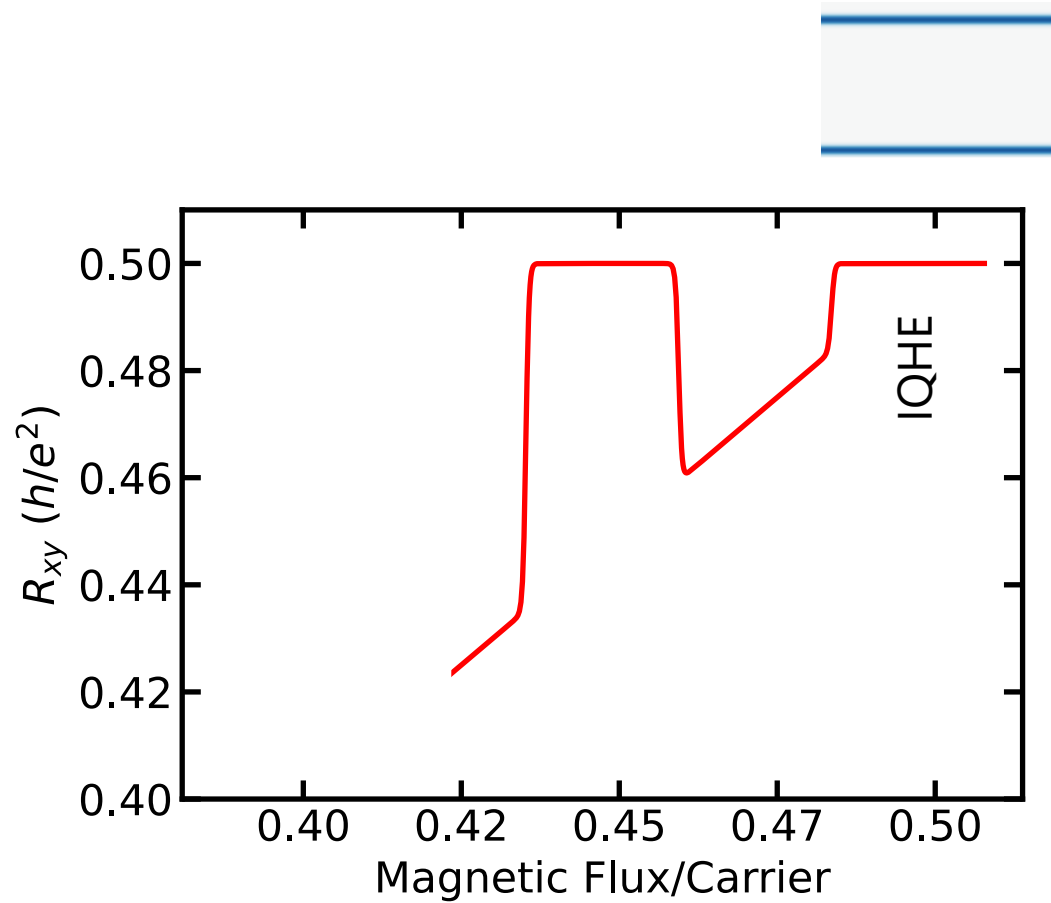
Charge ordered state:

- Wigner crystal, stripes, bubbles
- Reentrant quantum Hall

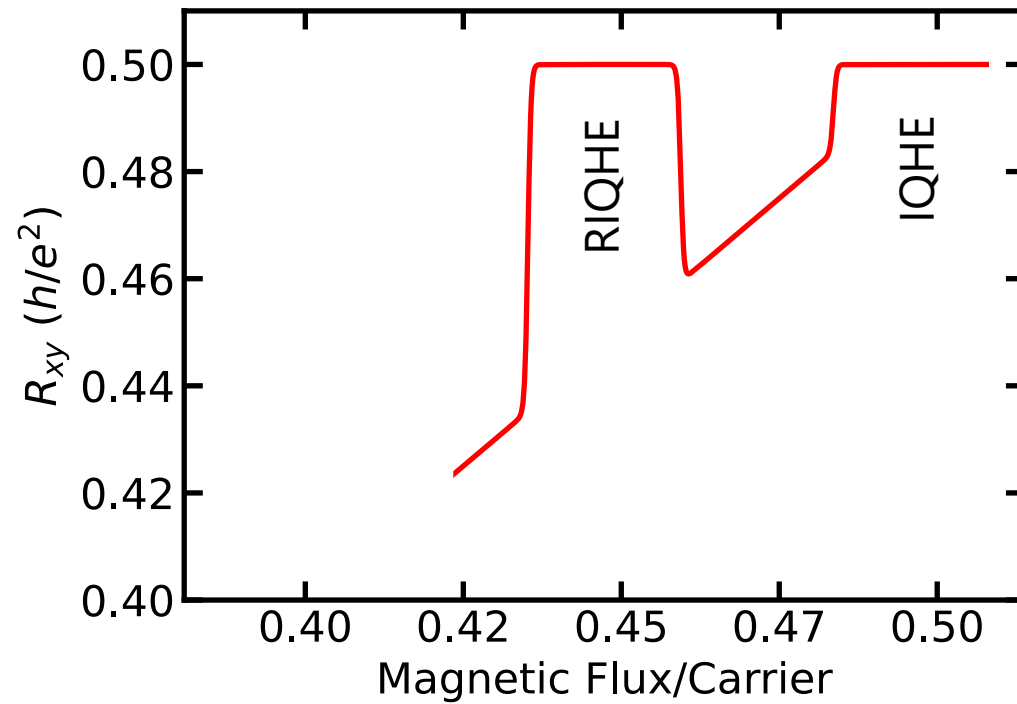
Reentrant Integer Quantum Hall (RIQHE)



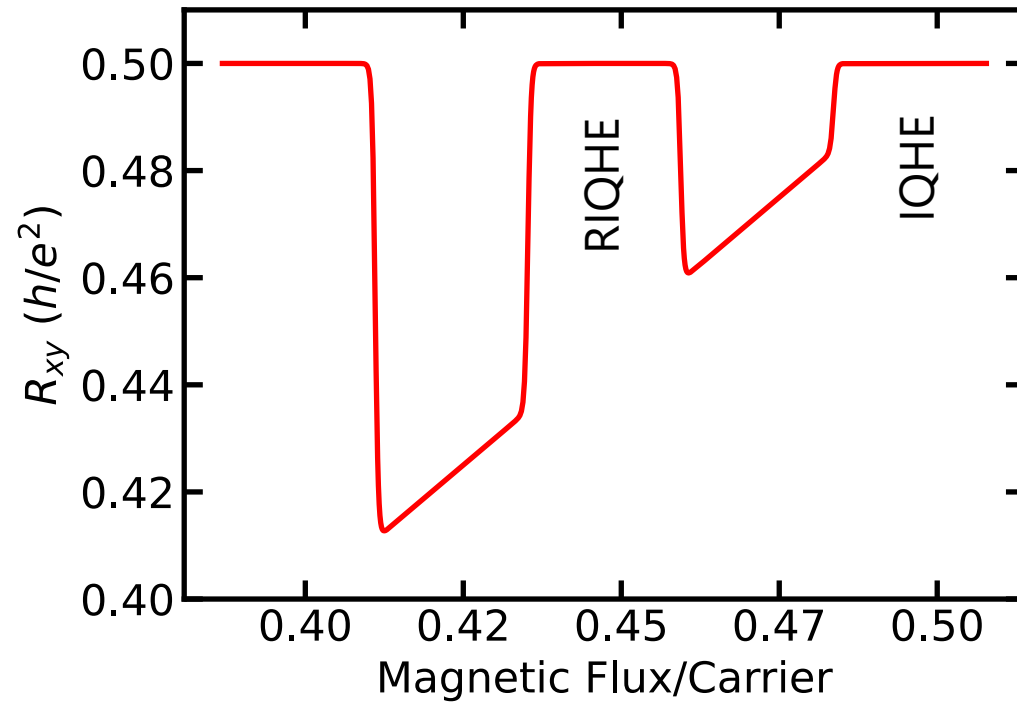
Reentrant Integer Quantum Hall (RIQHE)



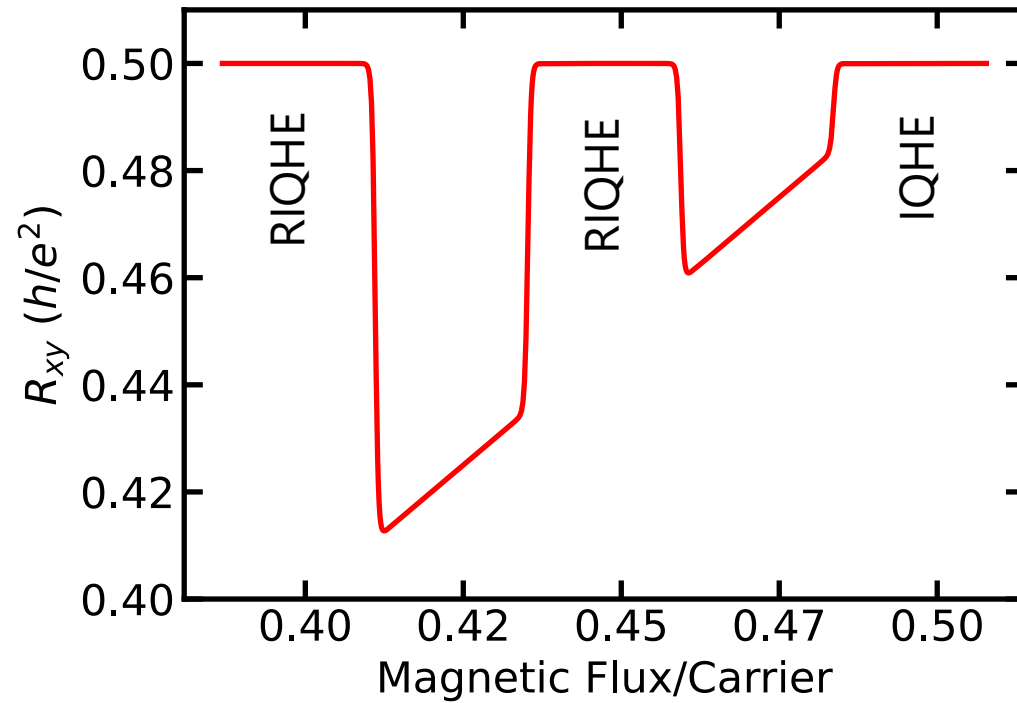
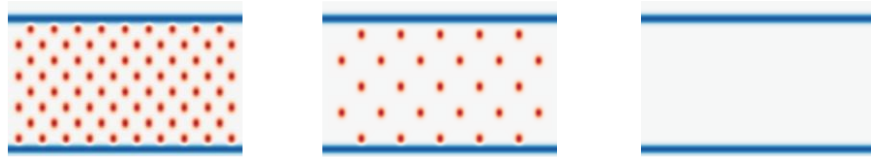
Reentrant Integer Quantum Hall (RIQHE)



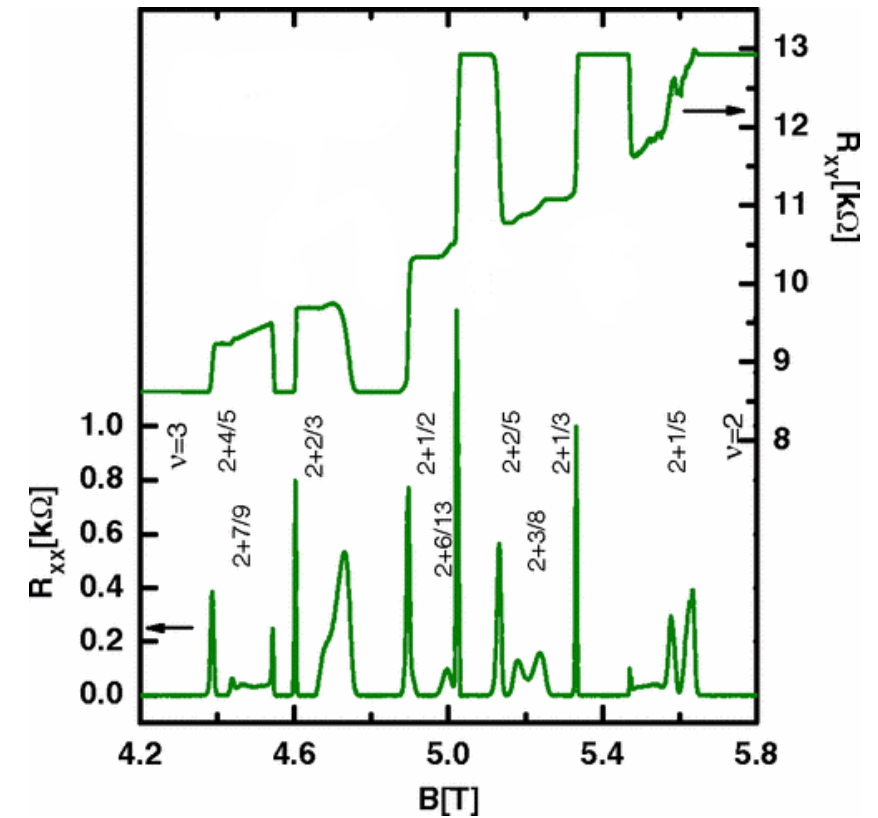
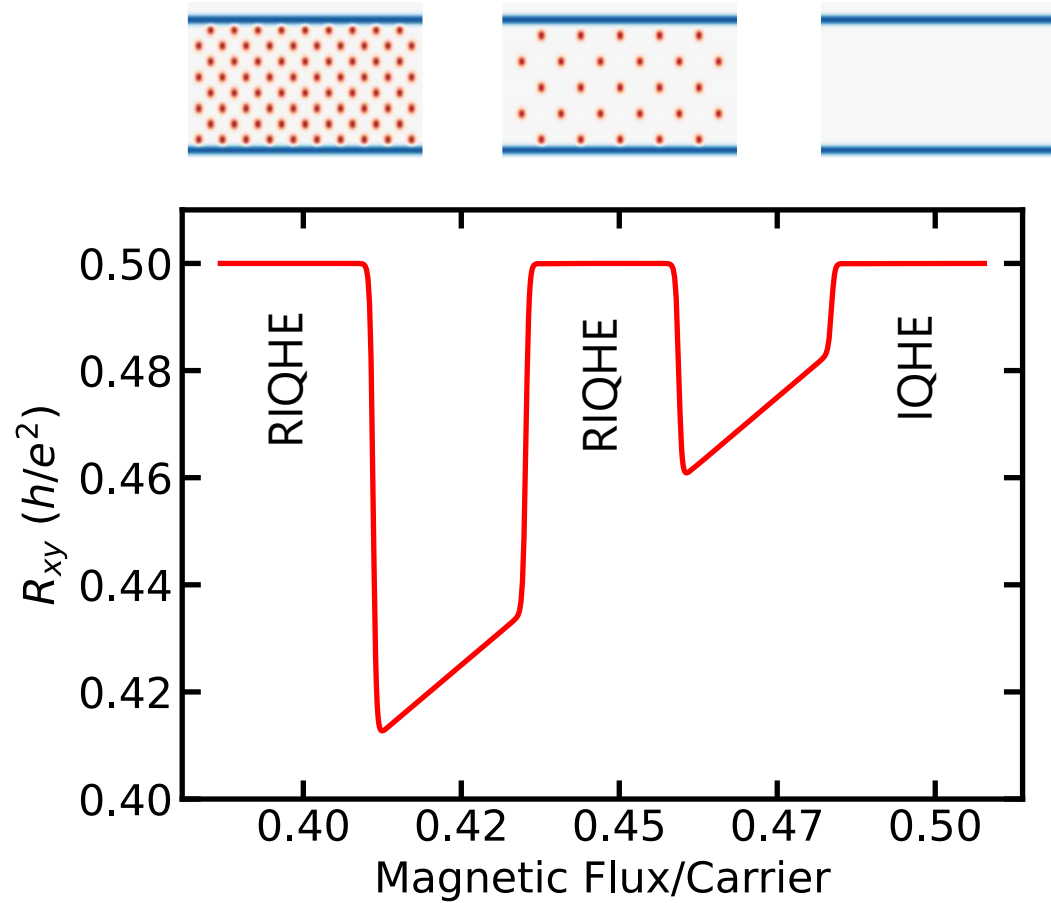
Reentrant Integer Quantum Hall (RIQHE)



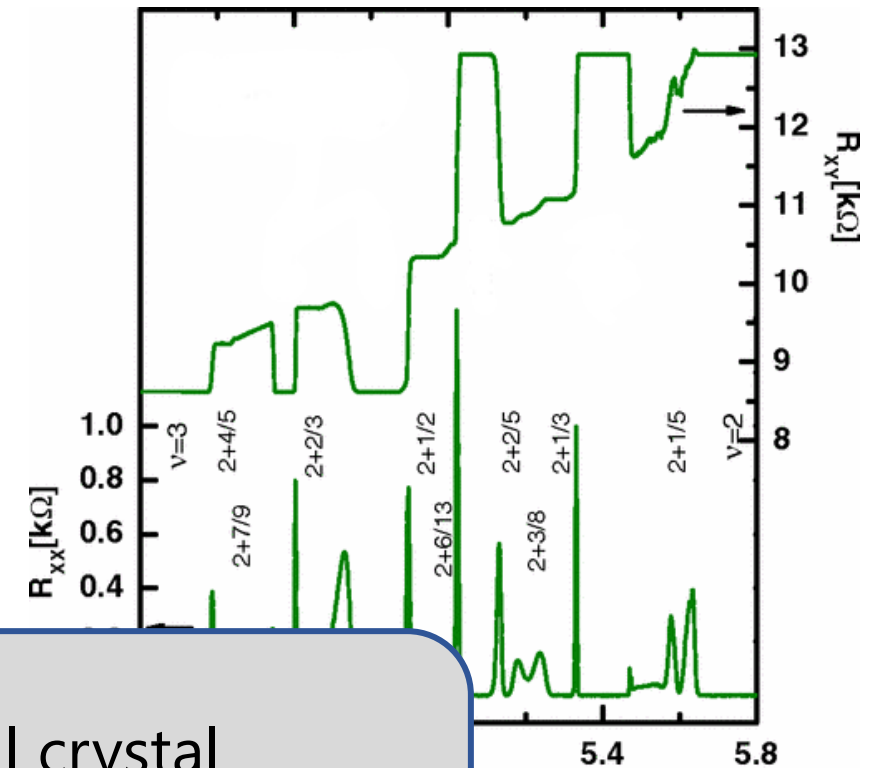
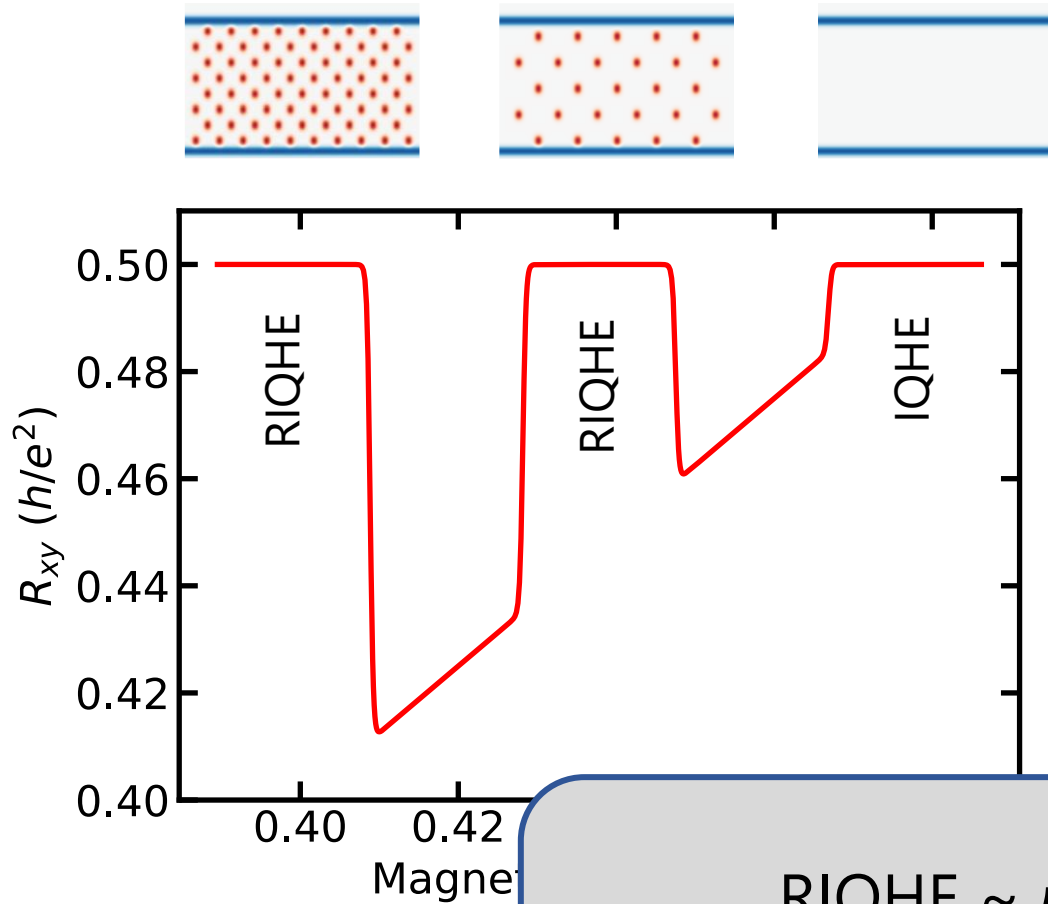
Reentrant Integer Quantum Hall (RIQHE)



Reentrant Integer Quantum Hall (RIQHE)

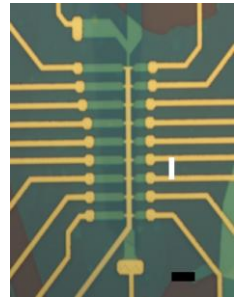


Reentrant Integer Quantum Hall (RIQHE)



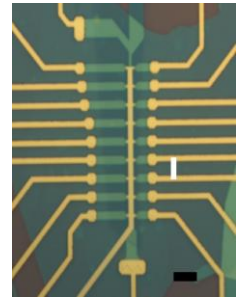
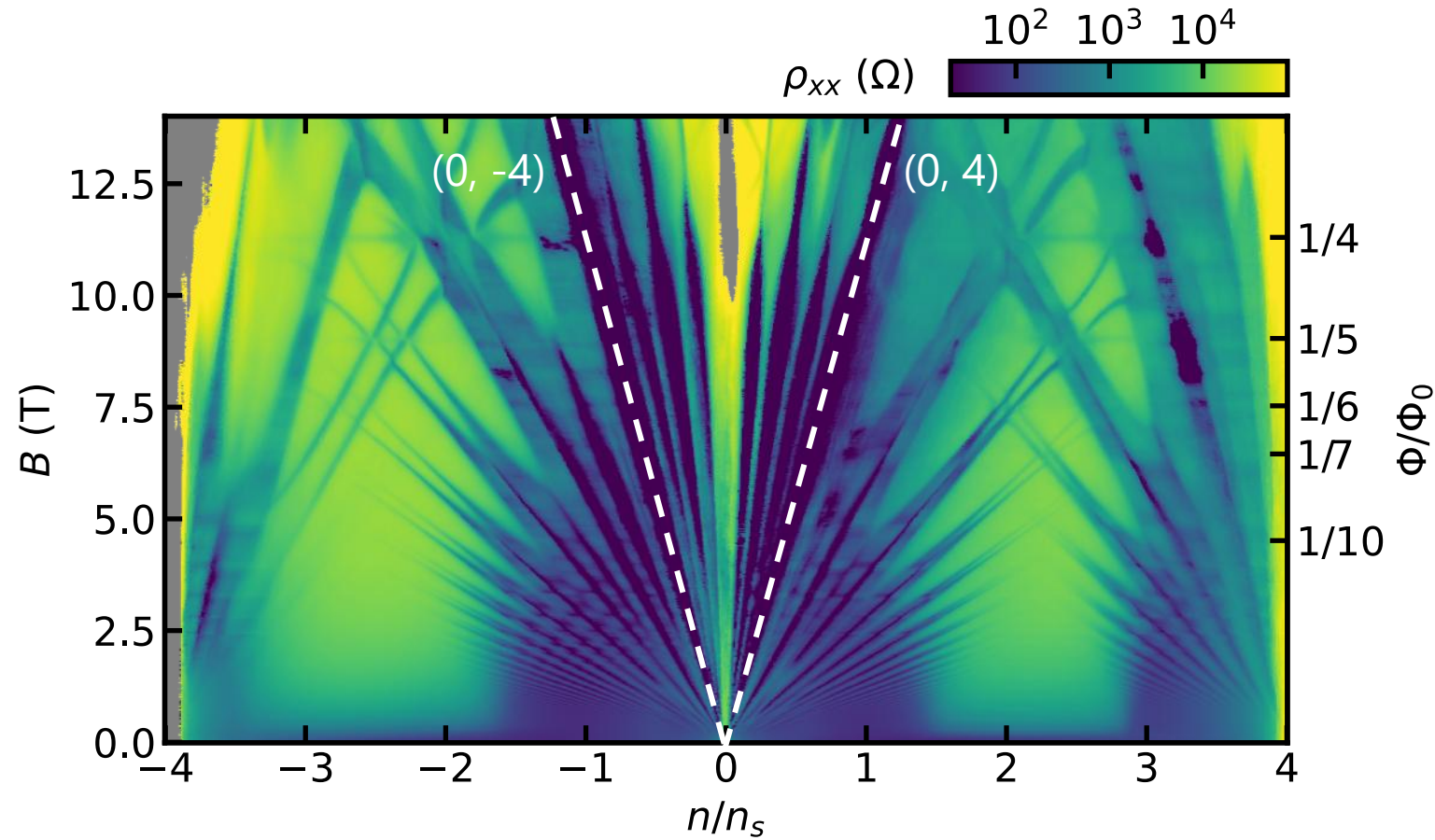
RIQHE ~ *partial* Hall crystal
What about *fractional* partial Hall crystals?
Crystallization when doping relative to a fractional gap

1.38° TBG: weak correlations at low fields



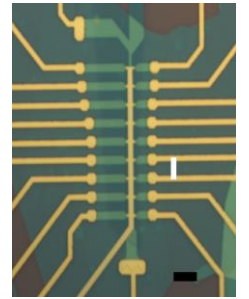
1.38° TBG

1.38° TBG: weak correlations at low fields

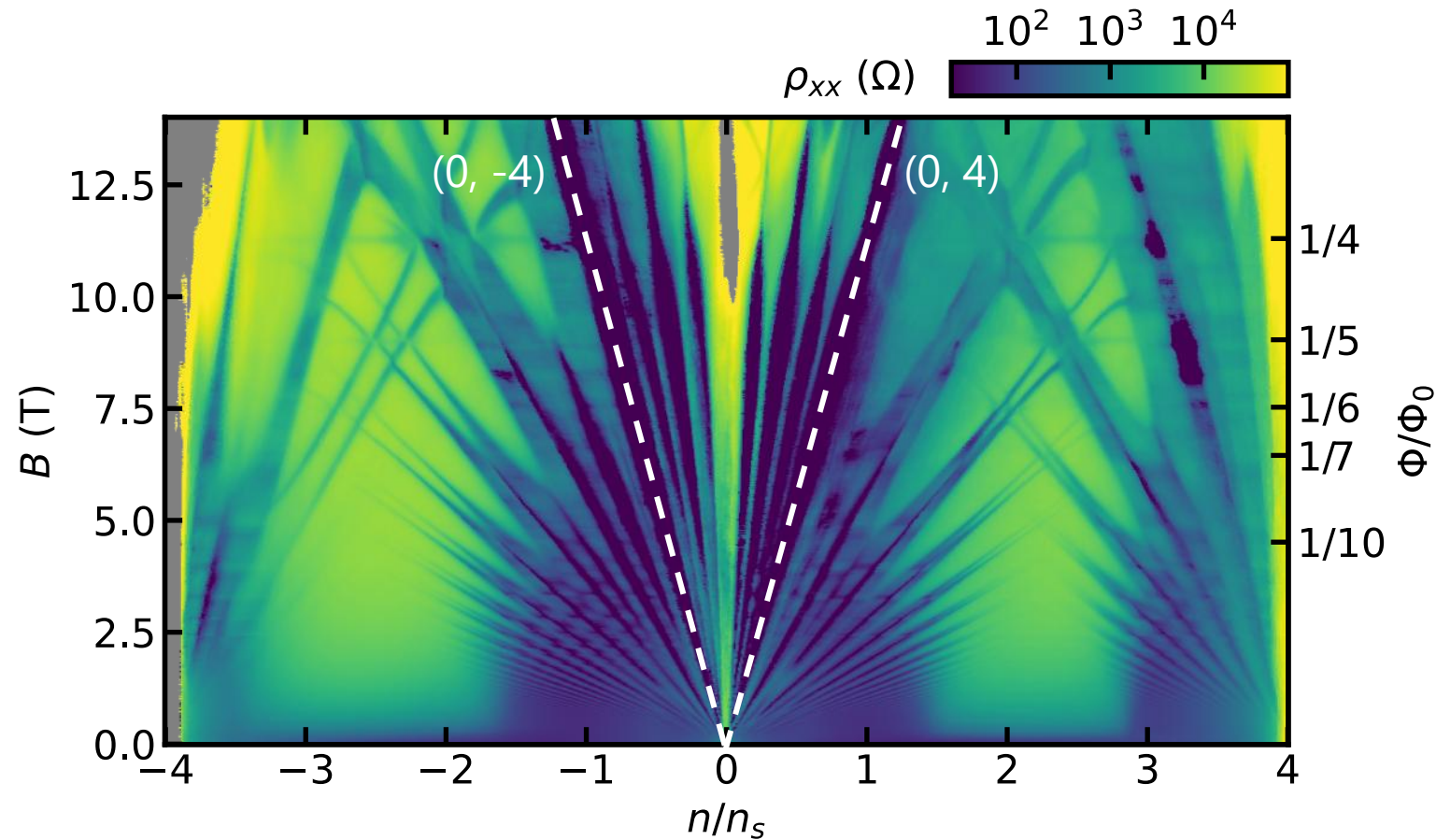


1.38° TBG

1.38° TBG: weak correlations at low fields



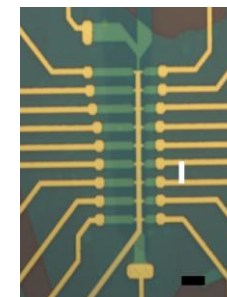
1.38° TBG



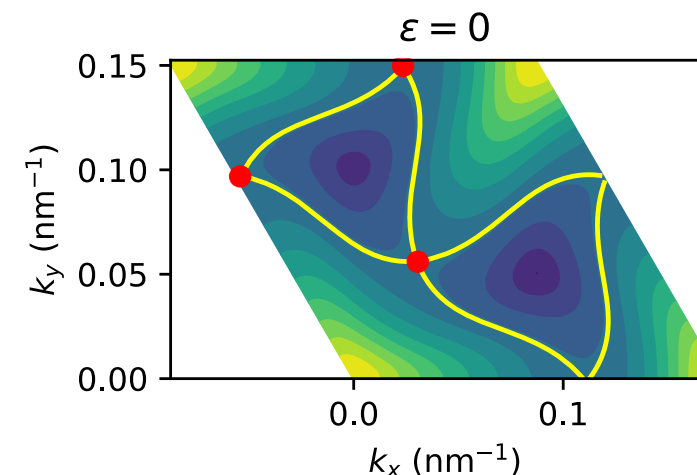
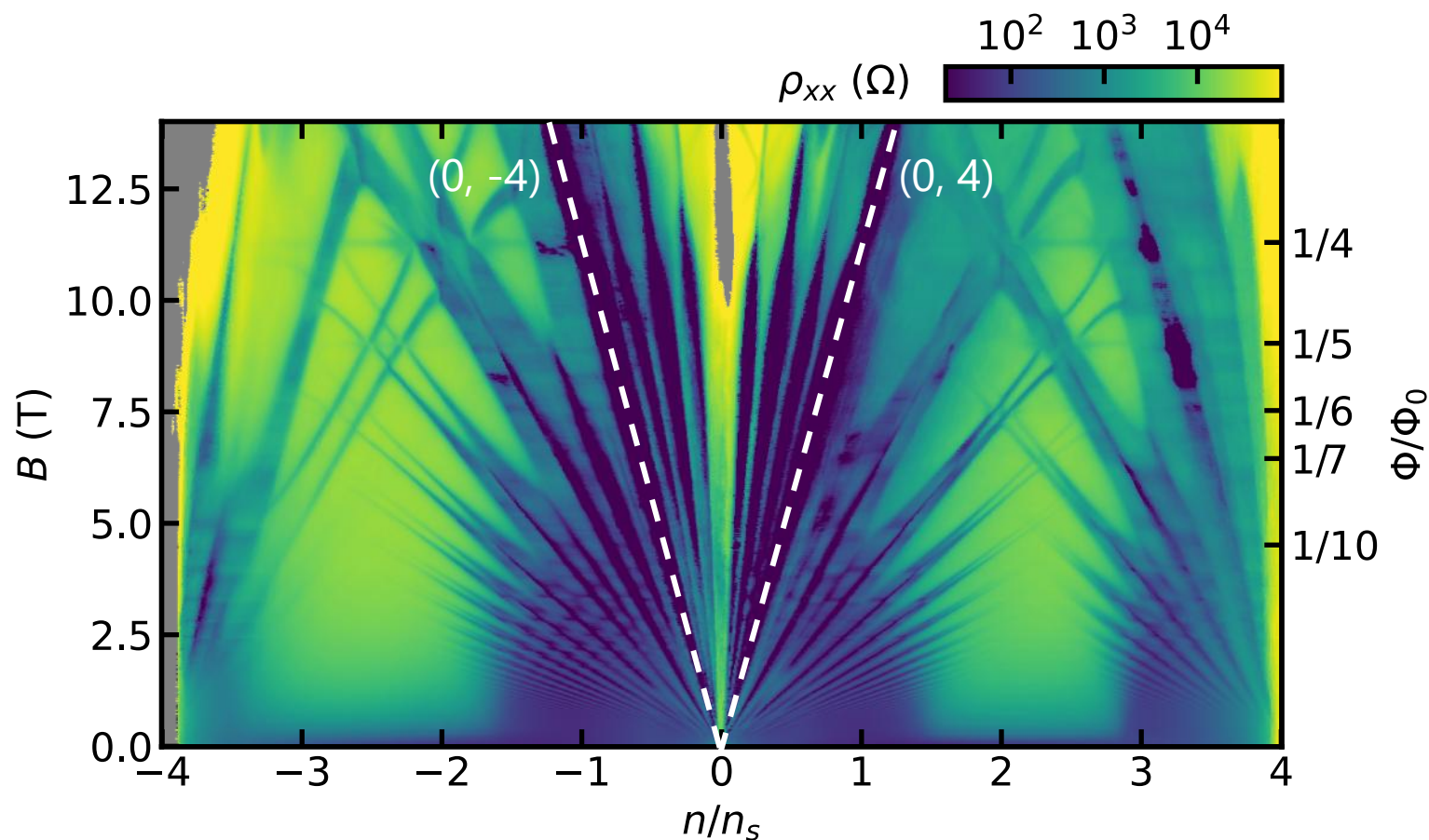
Diophantine: $n/n_s = s + t\Phi/\Phi_0$
Středa: $\sigma_{xy} = te^2/h$

Wang, Finney, Sharpe, et al. PNAS (2023)
Finney, Sharpe, et al. PNAS (2022)

1.38° TBG: weak correlations at low fields



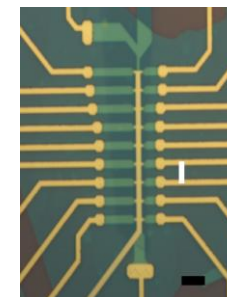
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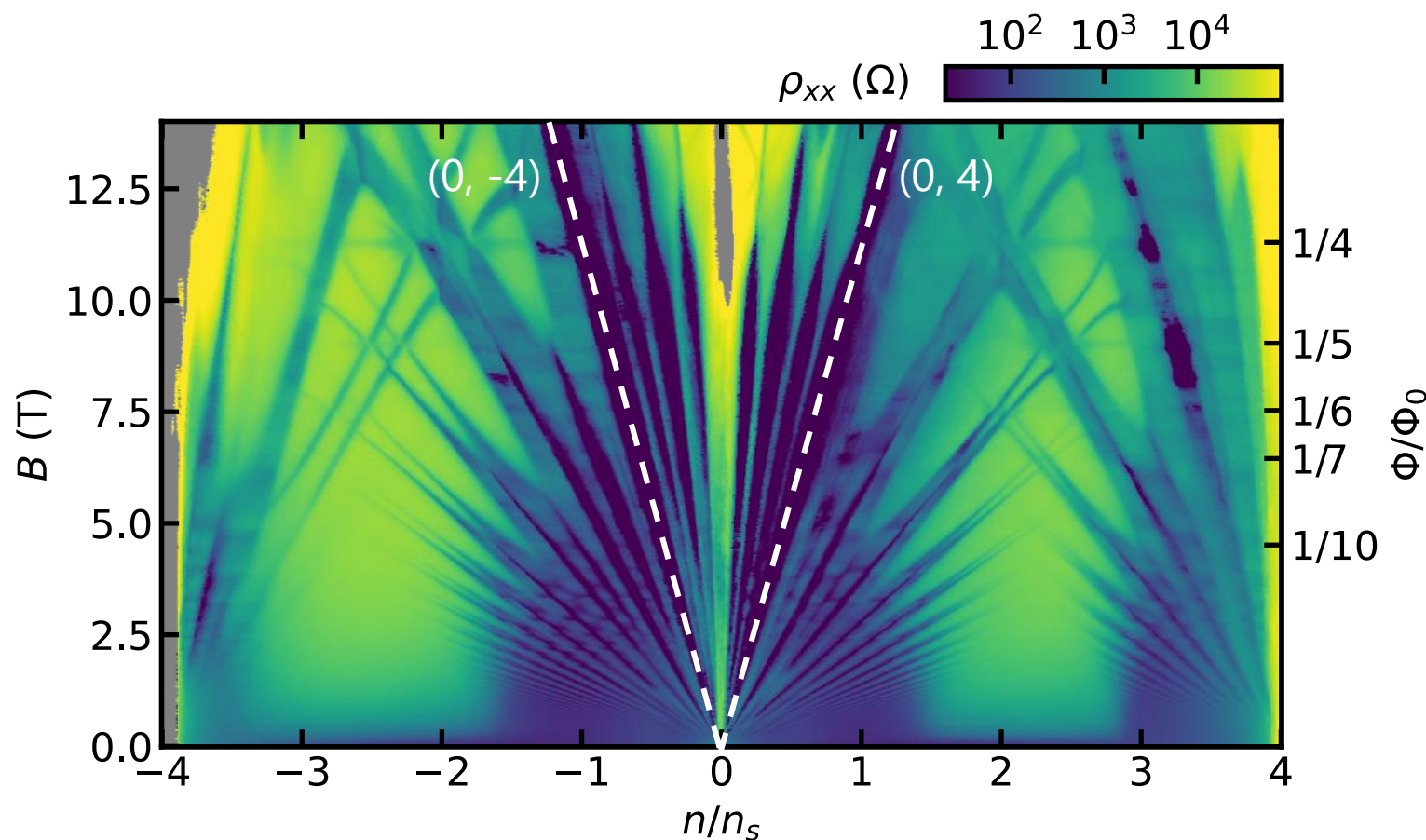
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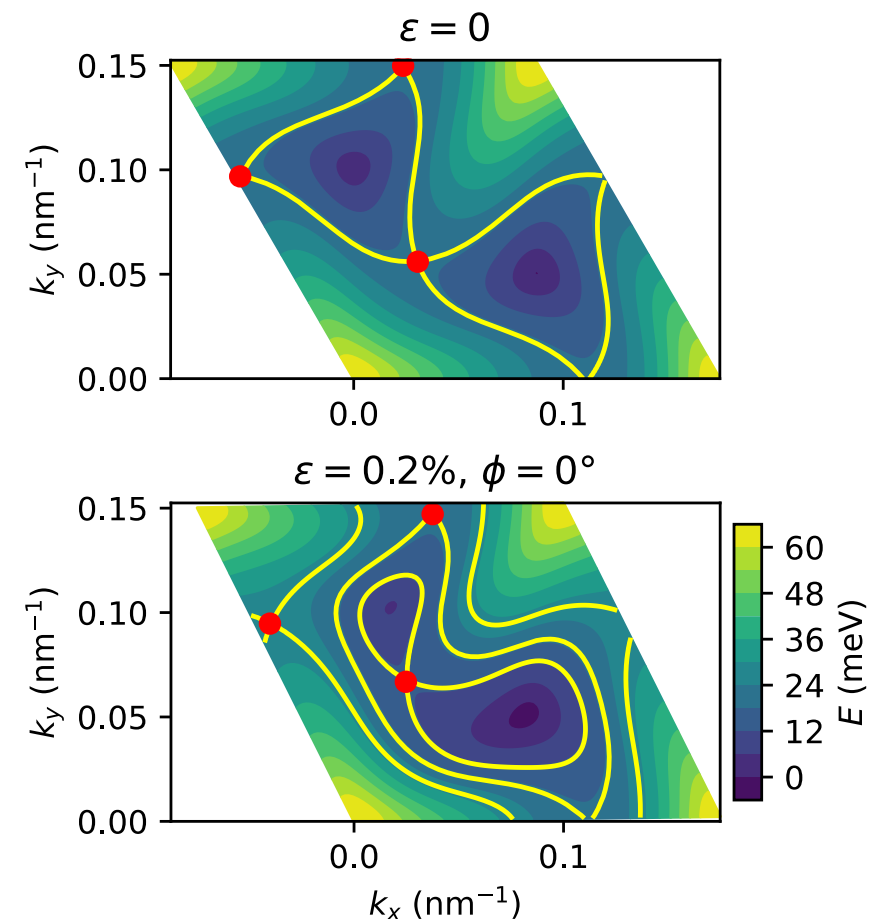
1.38° TBG: weak correlations at low fields



1.38° TBG

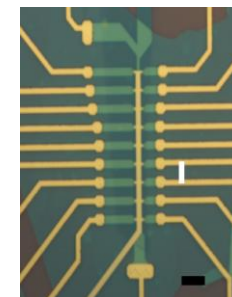
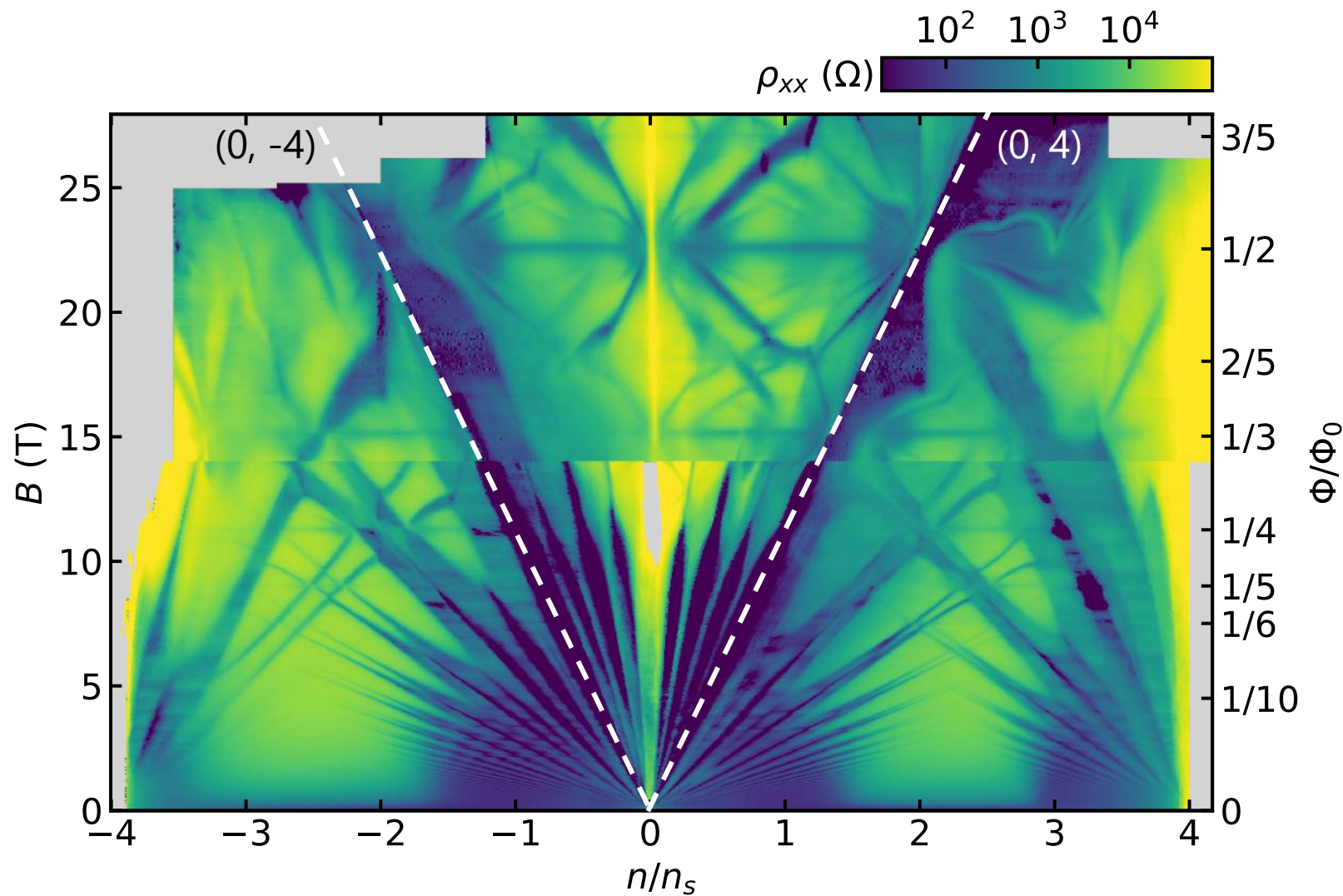


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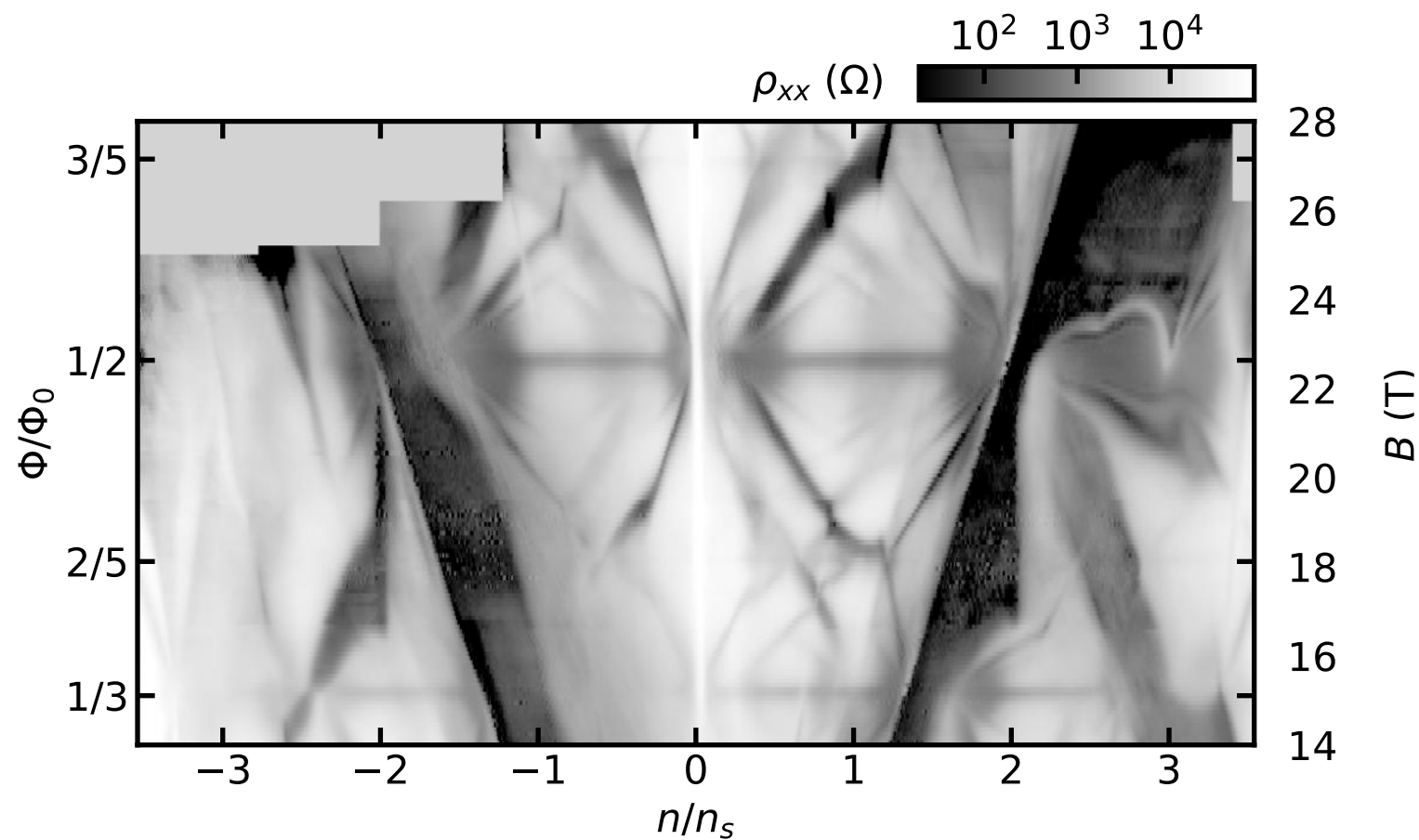
Wang, Finney, Sharpe, et al. PNAS (2023)
 Finney, Sharpe, et al. PNAS (2022)

Field induced correlation effects?



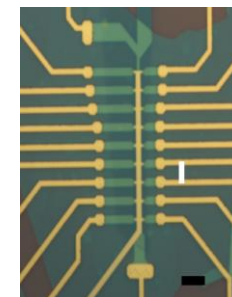
1.38° TBG

Fractional Chern insulators



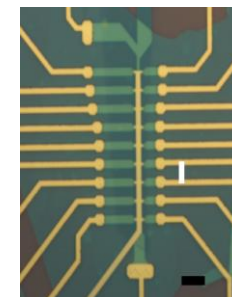
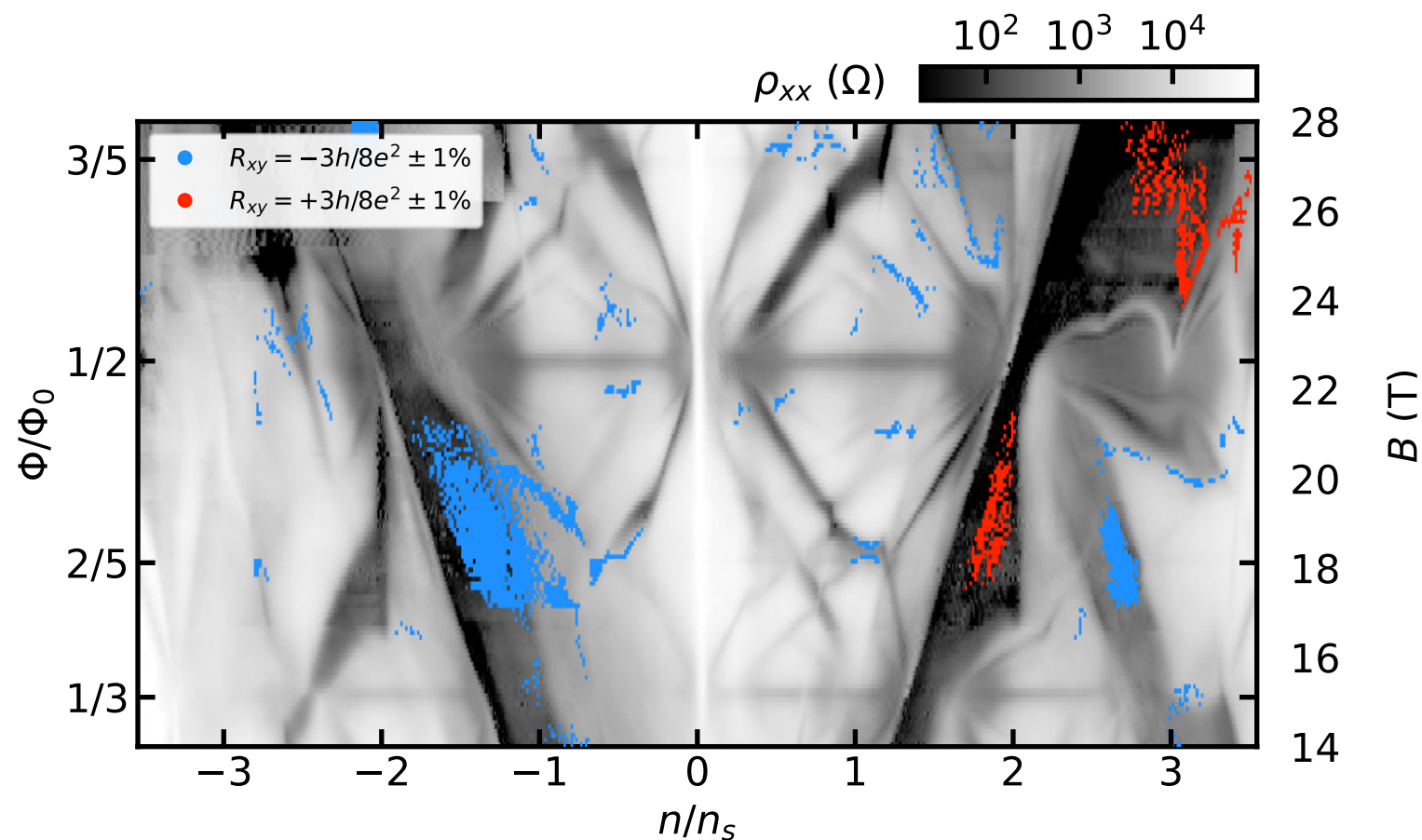
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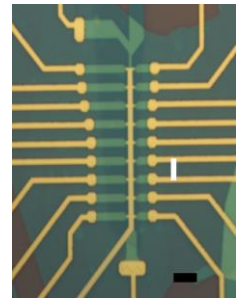
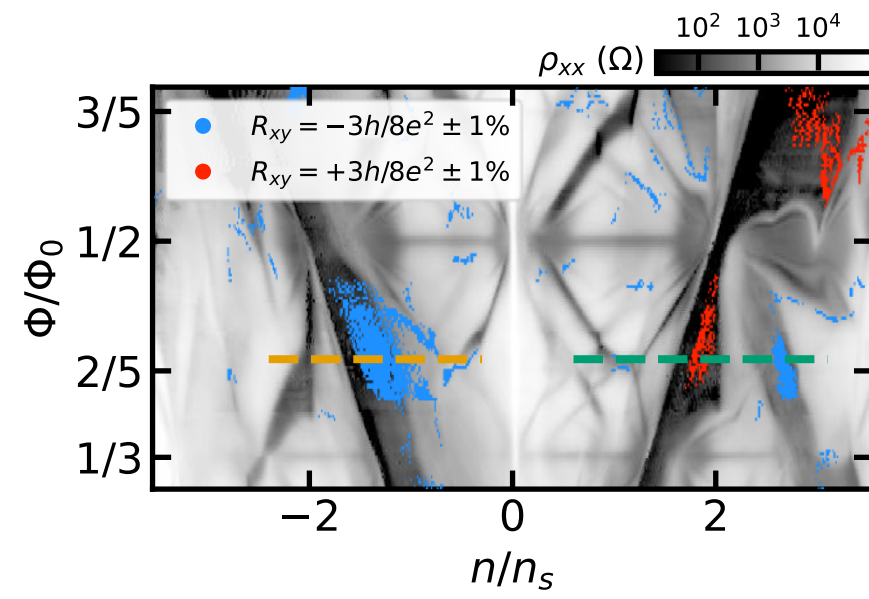
Fractional Chern insulators



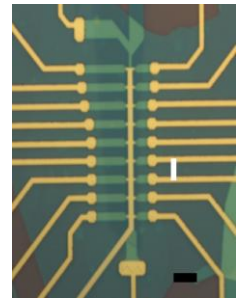
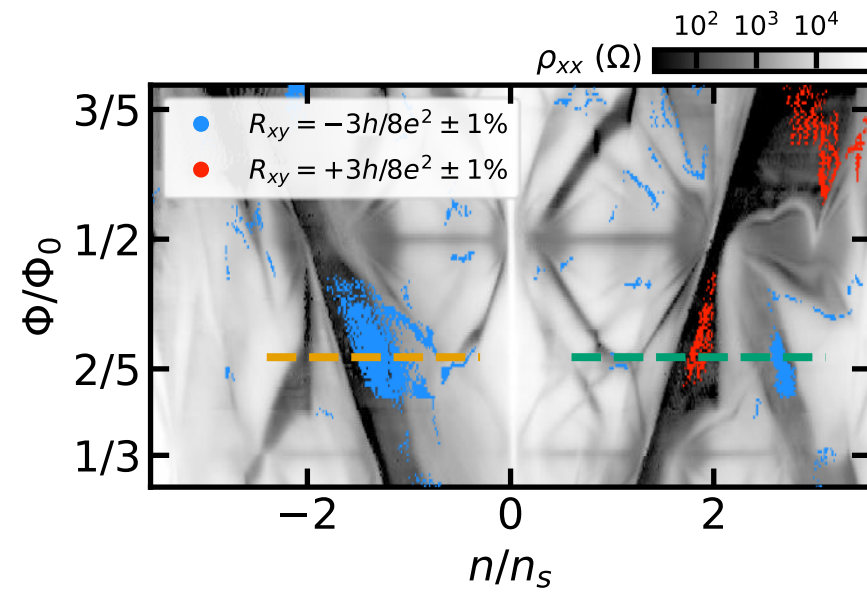
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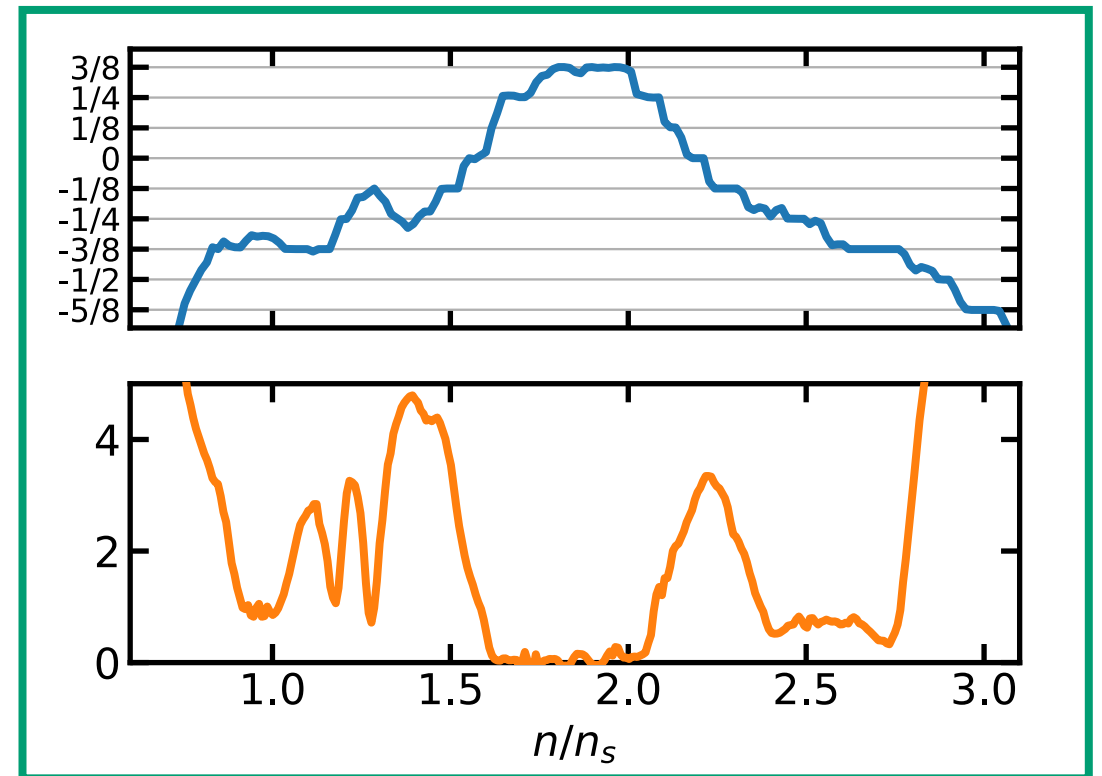
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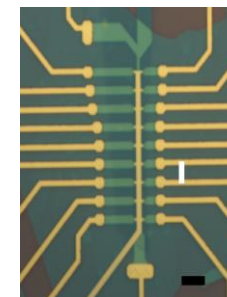
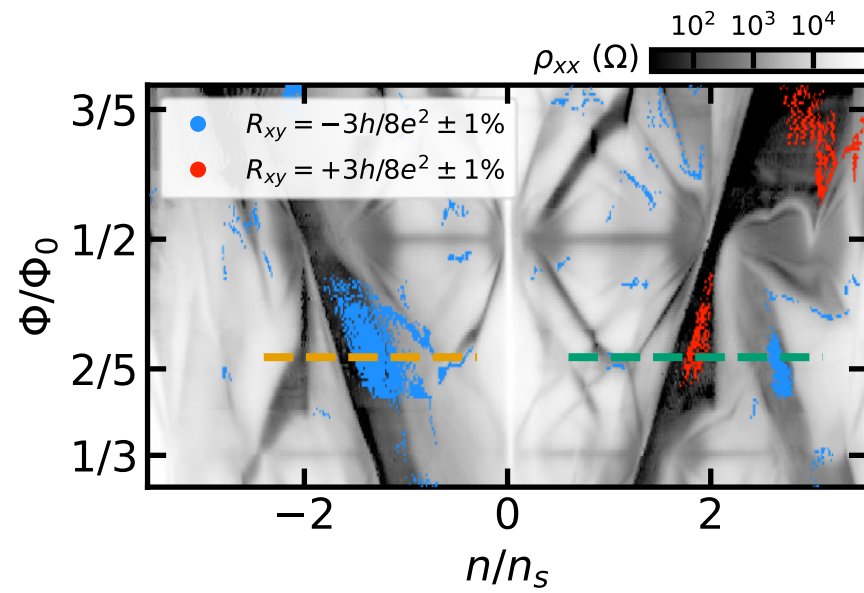


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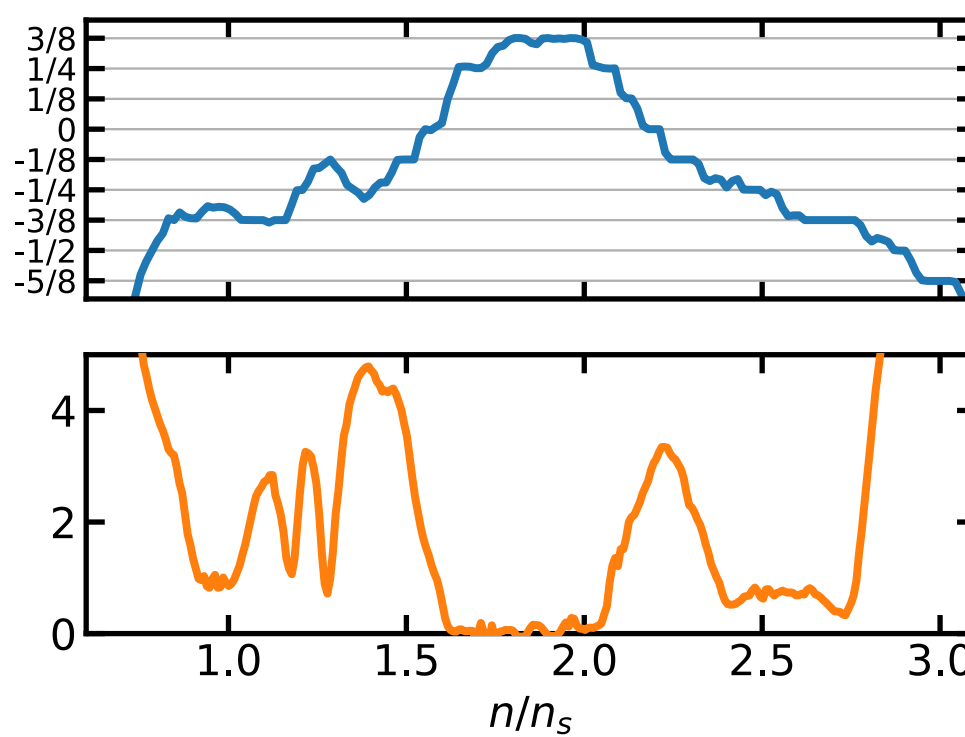
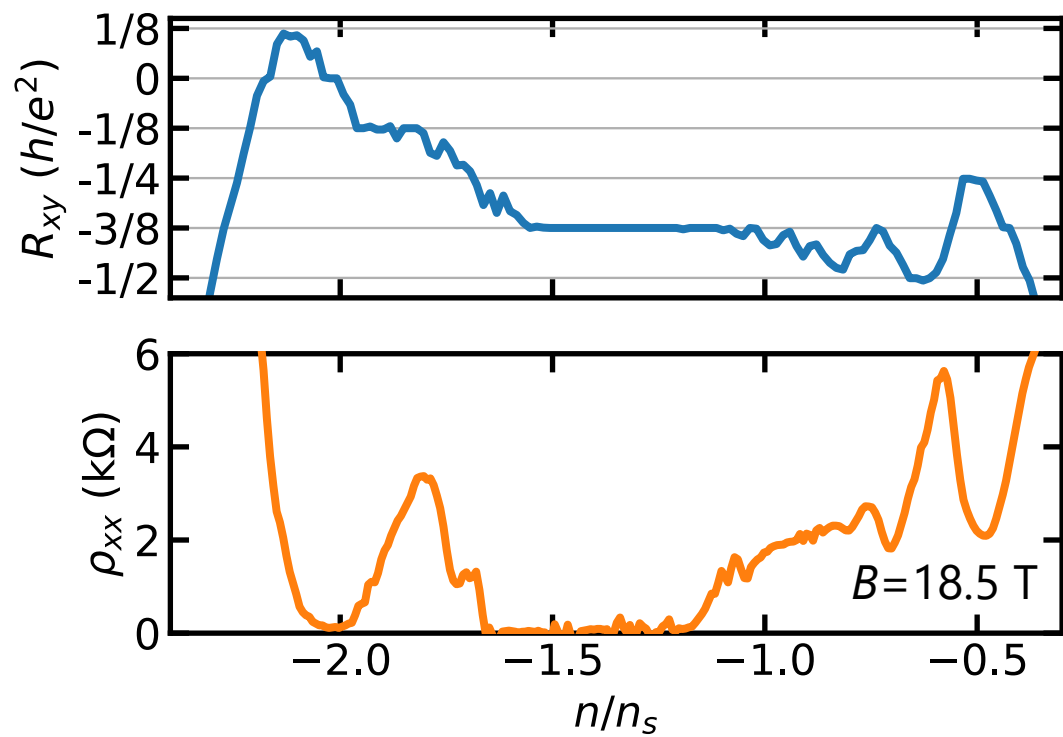


1.38° TBG

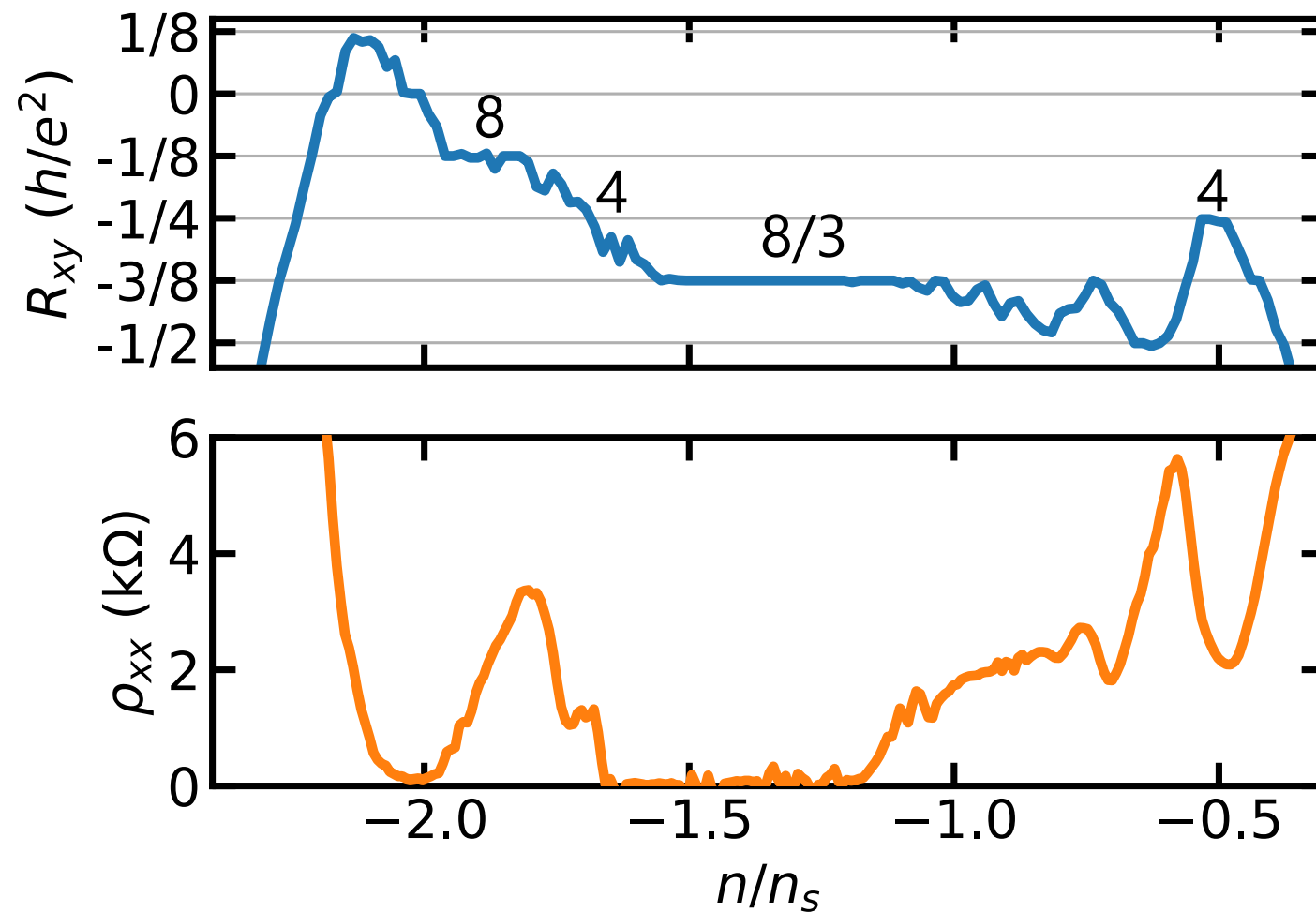




1.38° TBG



Extremely robust fractional state



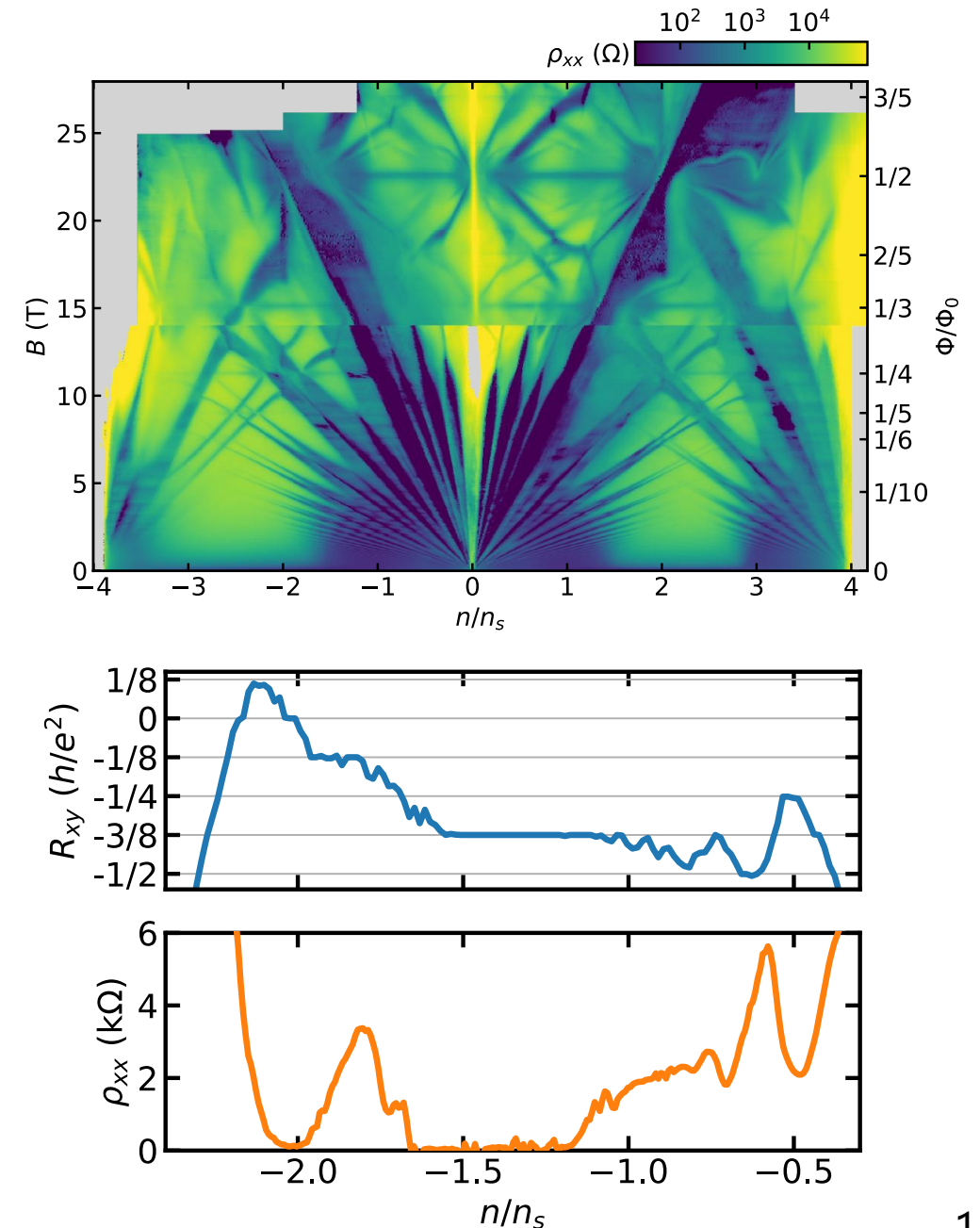
Summary

We still have much to learn about TBG!

Extended fractional Chern insulator near half flux!

Likely not extended because of **disorder**
More robustly quantized than any integer state

Fractional partial Hall crystal?



Acknowledgments



Joe Finney



Xiaoyu Wang



Jian Kang



David Goldhaber-Gordon



Marc Kastner



Oskar Vafek

Thanks

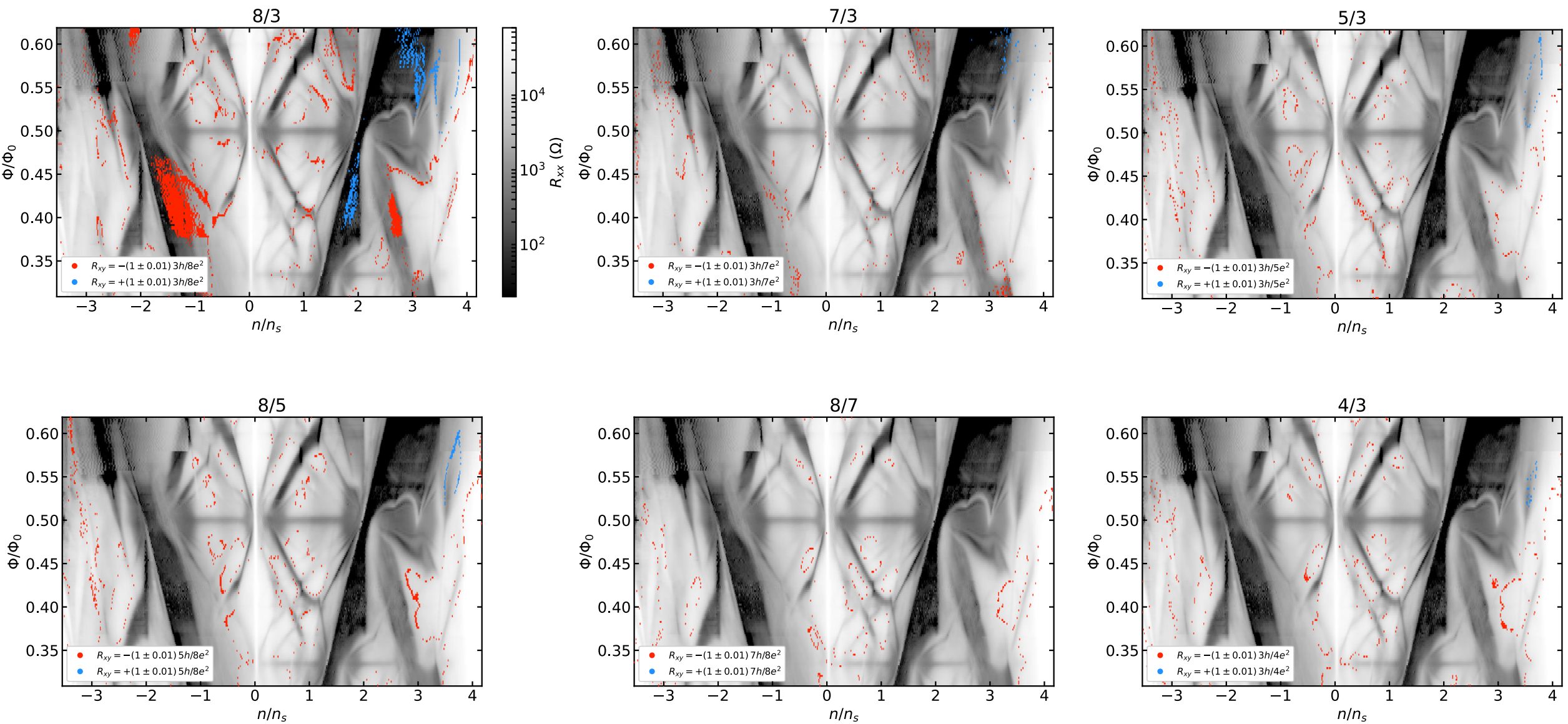
Linsey Rodenbach, Connie Hsueh

Takashi Taniguchi and Kenji Watanabe

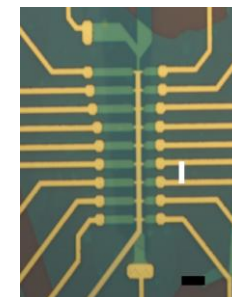
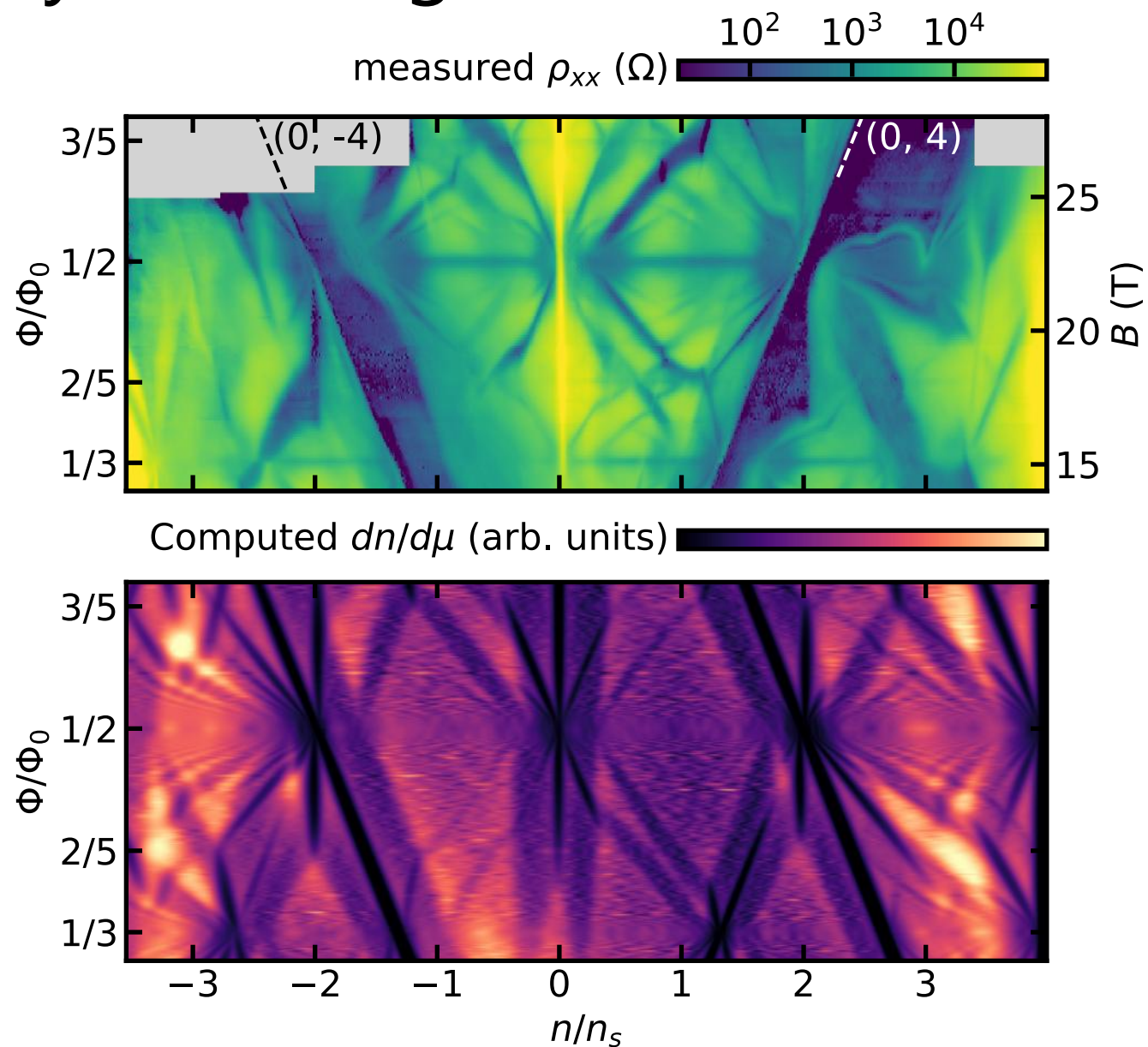
National High Magnetic Field Lab

Wonderful discussions with:

Steve Kivelson, Julian May-Mann, Trithep Devakul,
Yves Kwan, Tomohiro Soejima, Patrick Ledwith,
Ben Feldman, Matt Yankowitz, Jonah Herzon-Arbeitman,
Nisarga Paul, Sayak Bhattacharjee



Correlated physics at high field



1.38° TBG